

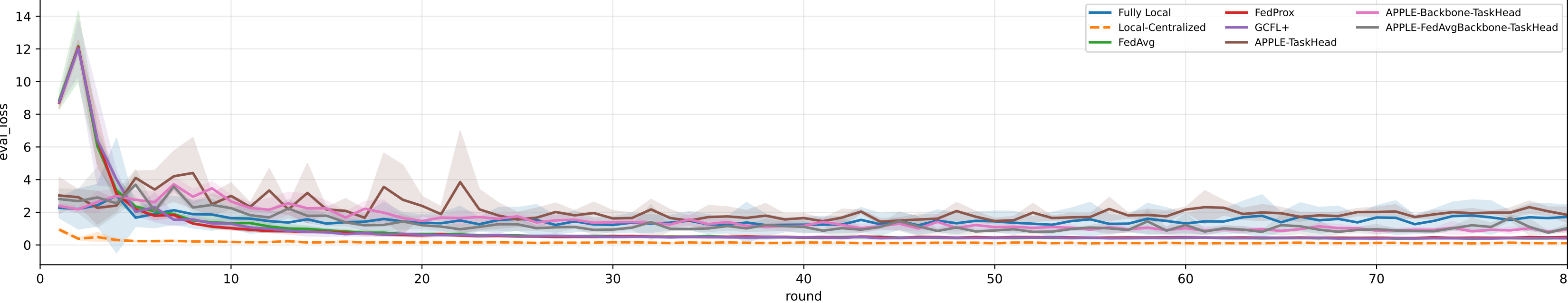
Controlled q-label heterogeneity: Mid | aggregated / pooled performance over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle2 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

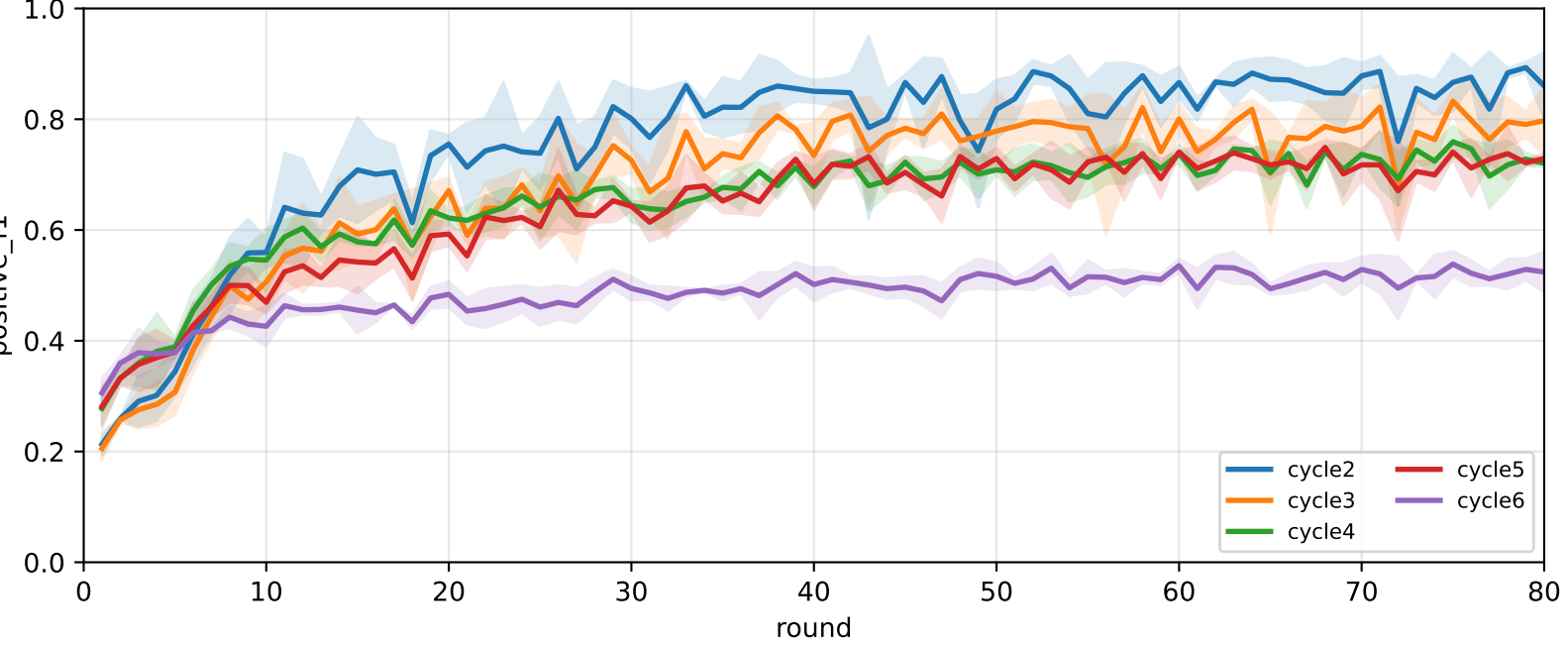
Best logged test performance (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	61.6% ± 4.0%	66.0% ± 4.3%	81.6% ± 7.0%	78.2% ± 5.1%	64.0% ± 3.6%	58.6% ± 5.2%	44.4% ± 3.0%
Local-Centralized	84.3% ± 1.6%	86.3% ± 1.4%	96.7% ± 1.1%	94.7% ± 0.6%	87.4% ± 4.2%	80.9% ± 2.2%	71.8% ± 3.9%
FedAvg	<u>70.3% ± 1.4%</u>	<u>72.8% ± 1.1%</u>	86.2% ± 4.2%	81.8% ± 3.9%	67.9% ± 1.8%	<u>70.1% ± 4.1%</u>	<u>58.1% ± 1.9%</u>
FedProx	68.9% ± 0.6%	71.9% ± 1.2%	<u>88.7% ± 3.2%</u>	79.0% ± 2.6%	<u>71.2% ± 3.5%</u>	64.2% ± 1.4%	56.4% ± 1.3%
GCFL+	69.6% ± 2.6%	71.9% ± 2.6%	83.4% ± 3.6%	<u>81.9% ± 3.7%</u>	68.0% ± 1.7%	68.1% ± 4.0%	58.0% ± 2.4%
APPLE-TaskHead	58.9% ± 1.7%	60.8% ± 1.6%	72.0% ± 4.9%	62.1% ± 2.1%	60.7% ± 2.3%	55.9% ± 3.2%	50.4% ± 2.3%
APPLE-Backbone-TaskHead	65.2% ± 1.9%	66.6% ± 2.2%	71.8% ± 7.2%	73.4% ± 5.1%	64.9% ± 0.3%	66.3% ± 5.4%	56.0% ± 2.2%
APPLE-FedAvgBackbone-TaskHead	68.7% ± 1.1%	71.8% ± 1.1%	88.2% ± 1.3%	76.1% ± 3.8%	70.5% ± 2.1%	66.9% ± 0.8%	55.1% ± 1.2%

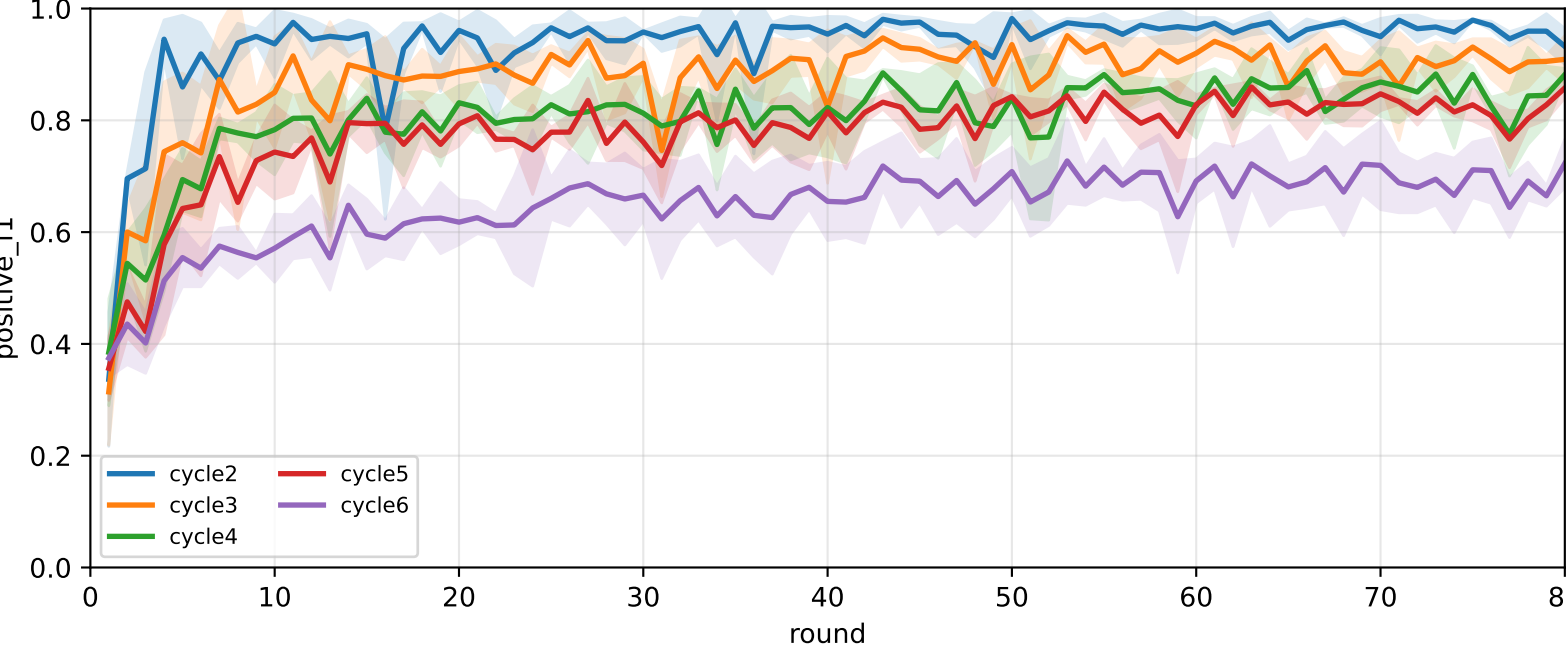
Validation loss comparison (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)



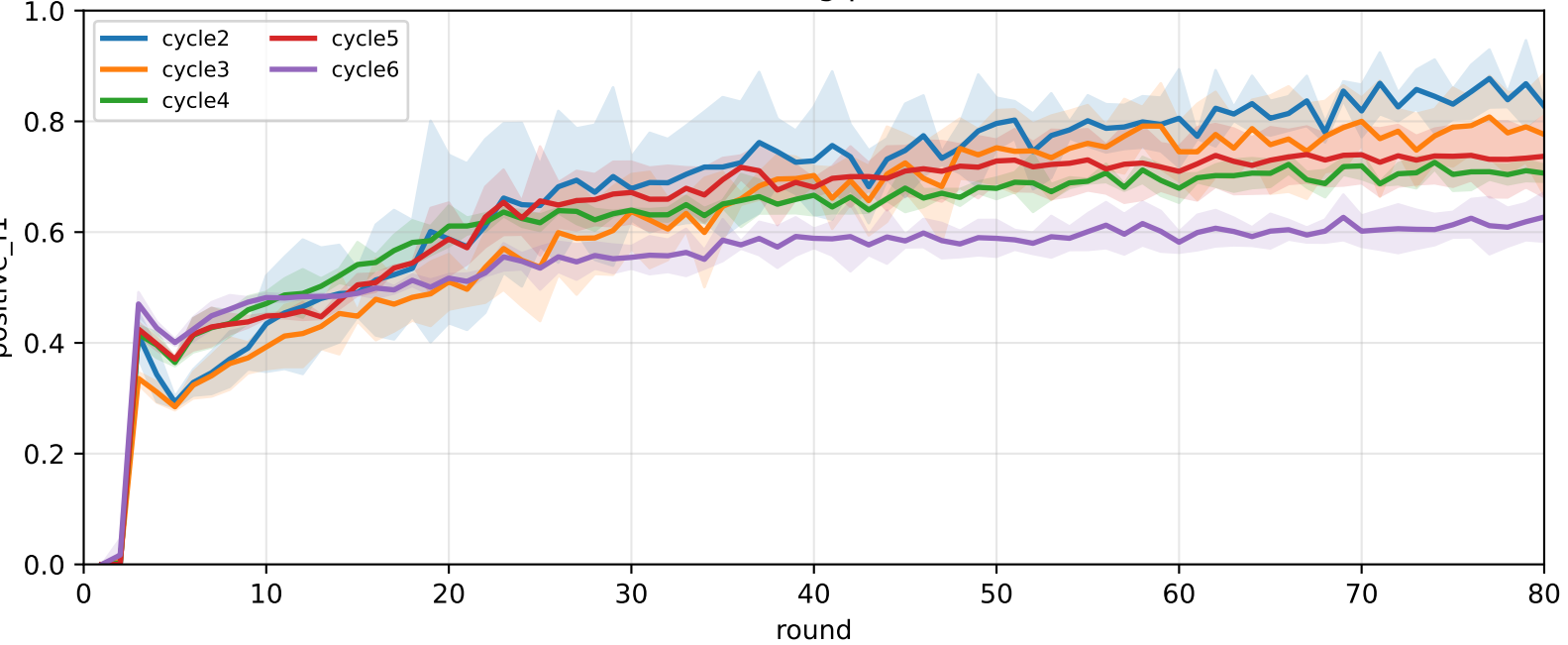
Pooled Fully Local positive F1



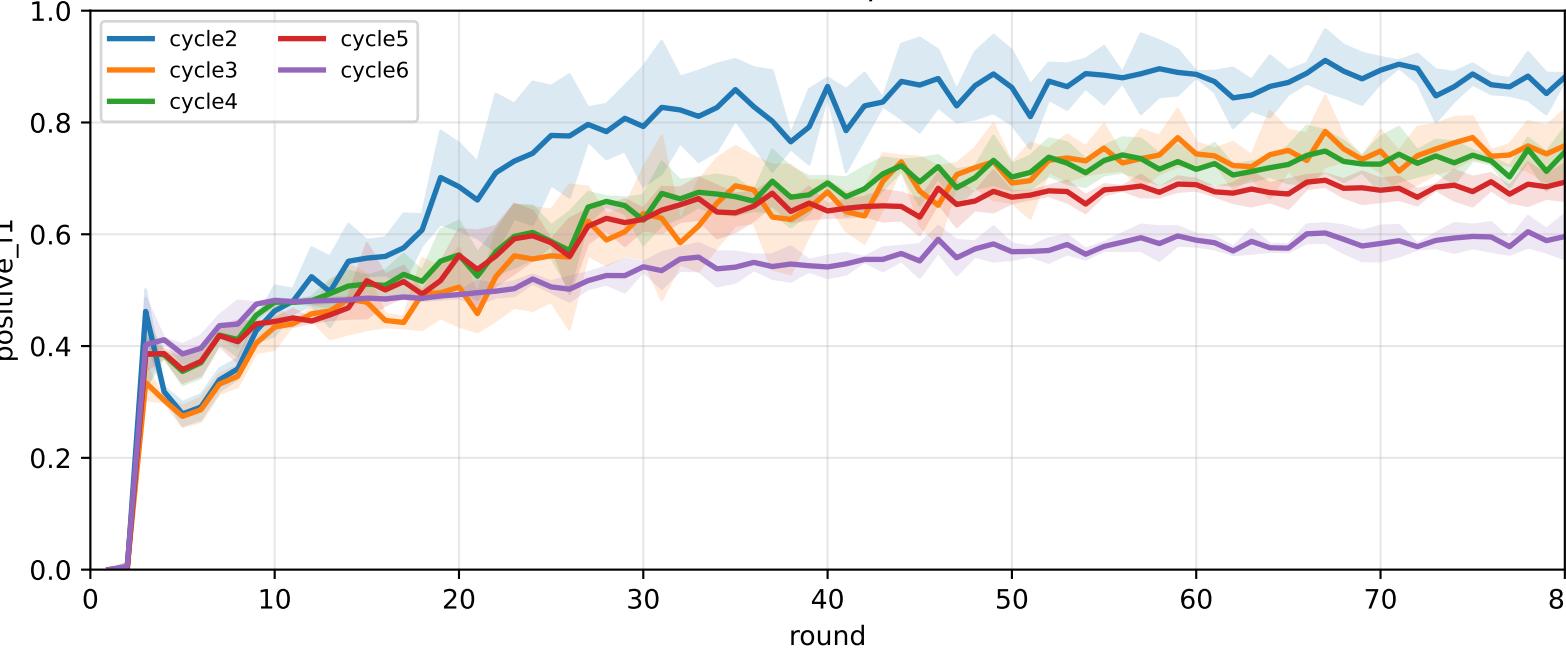
Pooled Local-Centralized positive F1



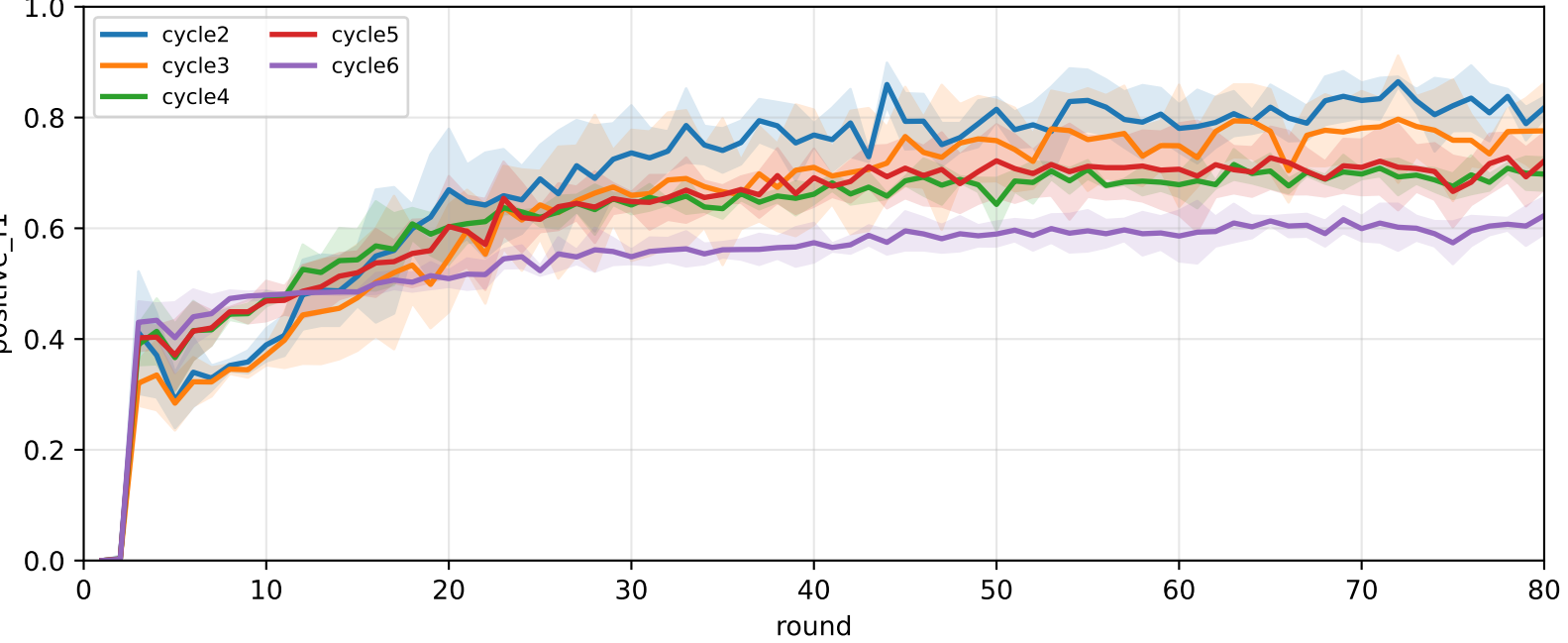
Pooled FedAvg positive F1



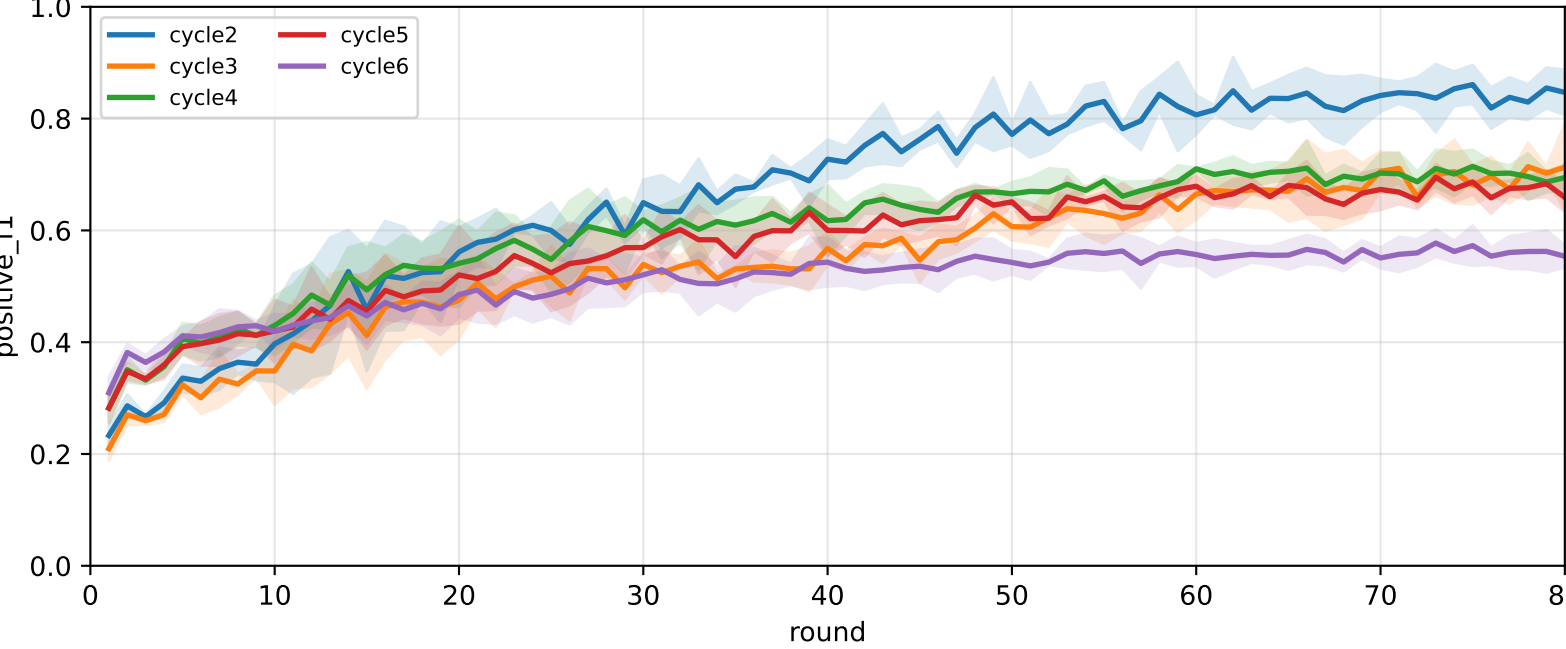
Pooled FedProx positive F1



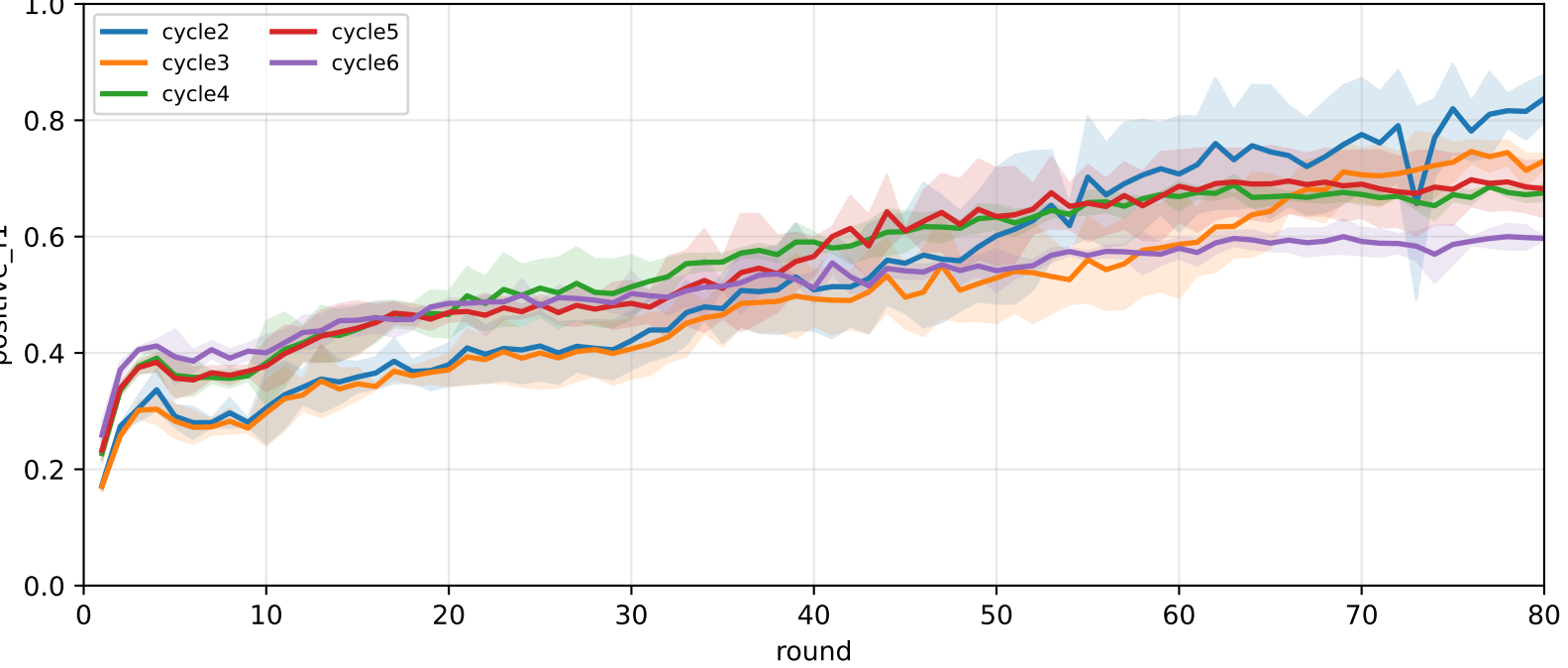
Pooled GCFL+ positive F1



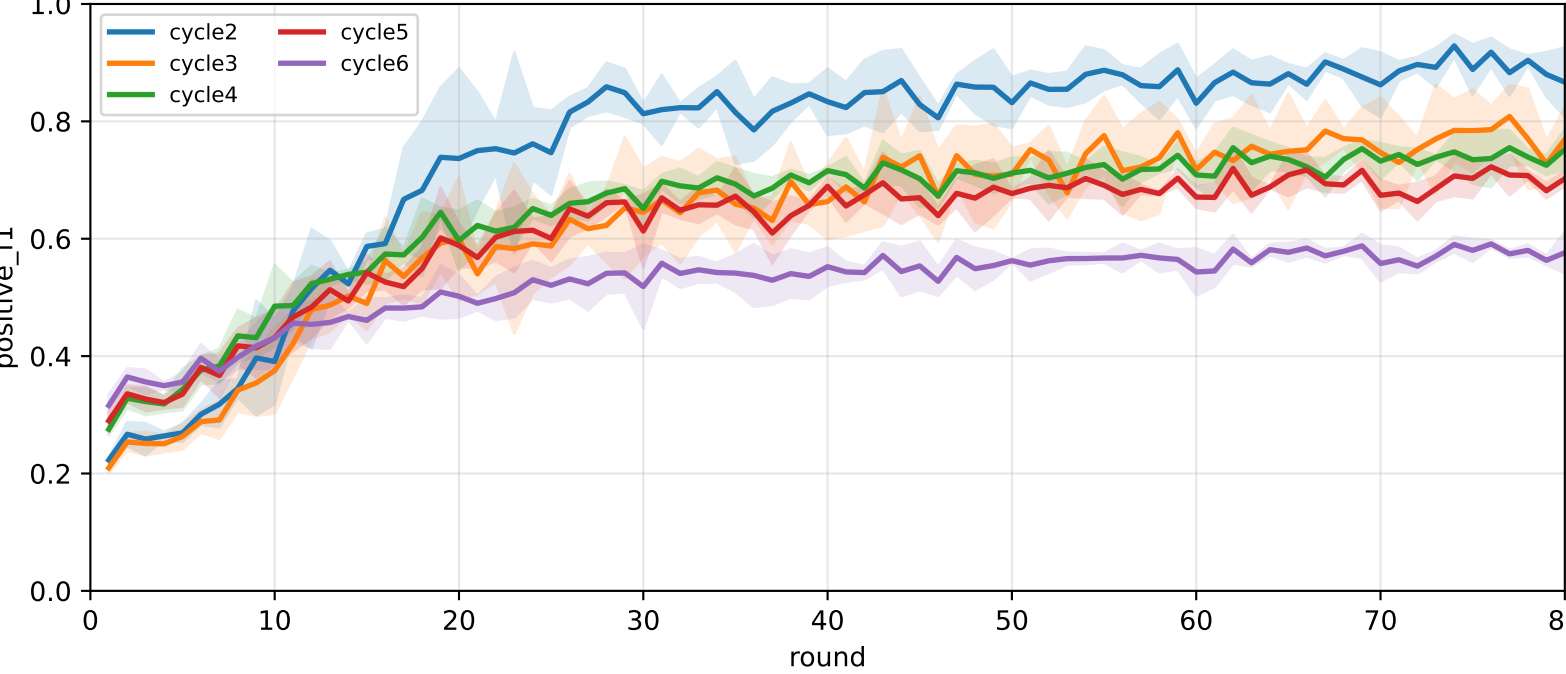
Pooled APPLE-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-Backbone-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-FedAvgBackbone-TaskHead positive F1 (after-local selected model, rounds=80)



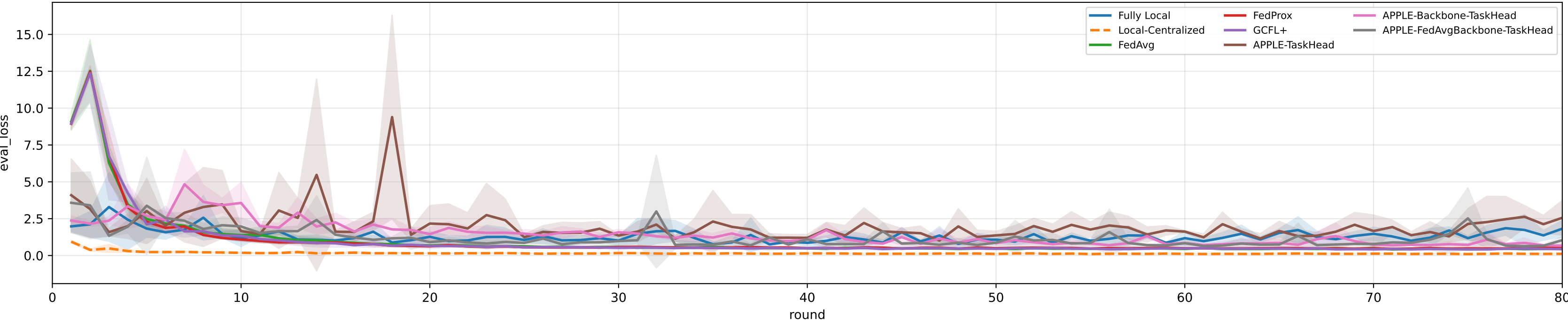
Controlled q-label heterogeneity: Mid | family aggregated / pooled performance |
family=q_task_cycle2 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle2 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% |
cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

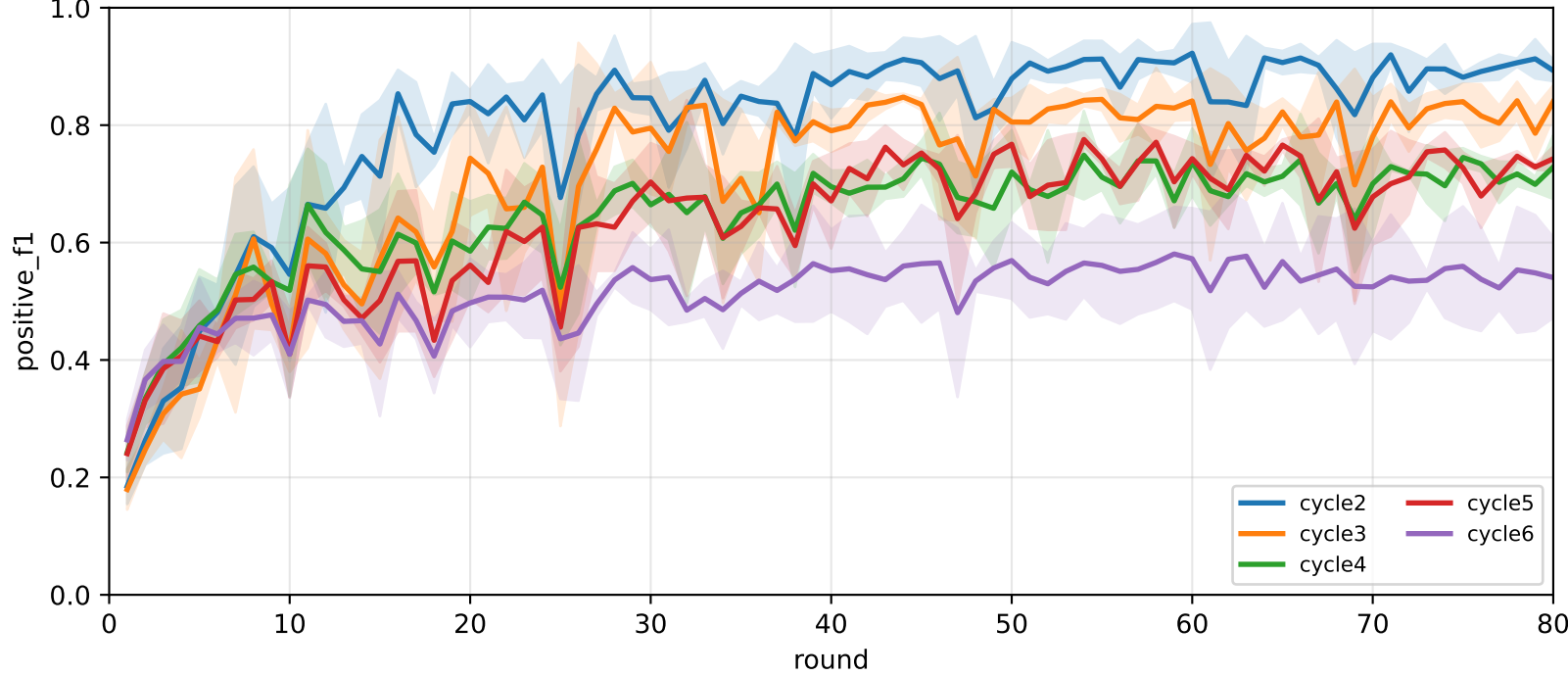
Best logged test performance (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	63.8% ± 5.4%	68.0% ± 5.8%	85.6% ± 8.0%	80.6% ± 4.3%	65.4% ± 6.2%	59.3% ± 9.8%	49.2% ± 2.7%
Local-Centralized	84.3% ± 1.6%	86.3% ± 1.4%	96.7% ± 1.1%	94.7% ± 0.6%	87.4% ± 4.2%	80.9% ± 2.2%	71.8% ± 3.9%
FedAvg	<u>70.2% ± 1.5%</u>	<u>72.6% ± 1.3%</u>	85.3% ± 3.3%	81.7% ± 3.7%	68.1% ± 1.3%	<u>69.9% ± 4.0%</u>	58.1% ± 2.0%
FedProx	68.8% ± 1.0%	71.6% ± 1.5%	87.8% ± 2.8%	78.3% ± 3.2%	<u>71.6% ± 4.1%</u>	64.0% ± 1.1%	56.4% ± 1.1%
GCFL+	69.9% ± 2.4%	72.1% ± 2.6%	83.7% ± 5.0%	<u>81.8% ± 3.5%</u>	68.5% ± 1.6%	68.5% ± 3.4%	<u>58.2% ± 1.8%</u>
APPLE-TaskHead	58.0% ± 1.5%	60.3% ± 1.5%	75.7% ± 6.0%	64.2% ± 1.7%	59.0% ± 3.9%	52.7% ± 4.0%	49.8% ± 1.7%
APPLE-Backbone-TaskHead	67.5% ± 1.0%	69.5% ± 0.1%	77.9% ± 8.9%	78.1% ± 2.0%	67.0% ± 2.8%	67.4% ± 5.9%	57.2% ± 3.1%
APPLE-FedAvgBackbone-TaskHead	68.2% ± 3.7%	71.6% ± 3.6%	<u>90.0% ± 6.5%</u>	80.2% ± 5.3%	69.2% ± 7.7%	63.2% ± 3.5%	55.4% ± 2.6%

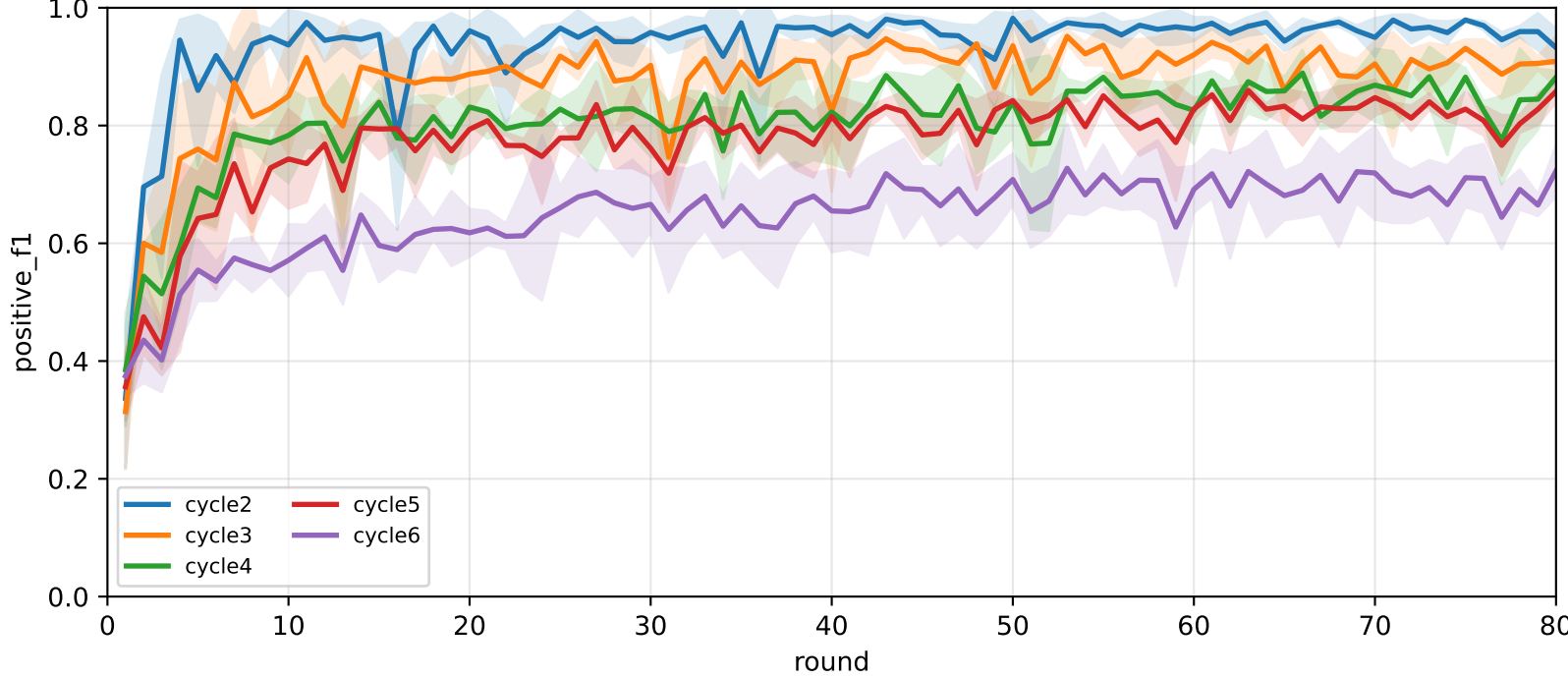
Validation loss comparison (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)



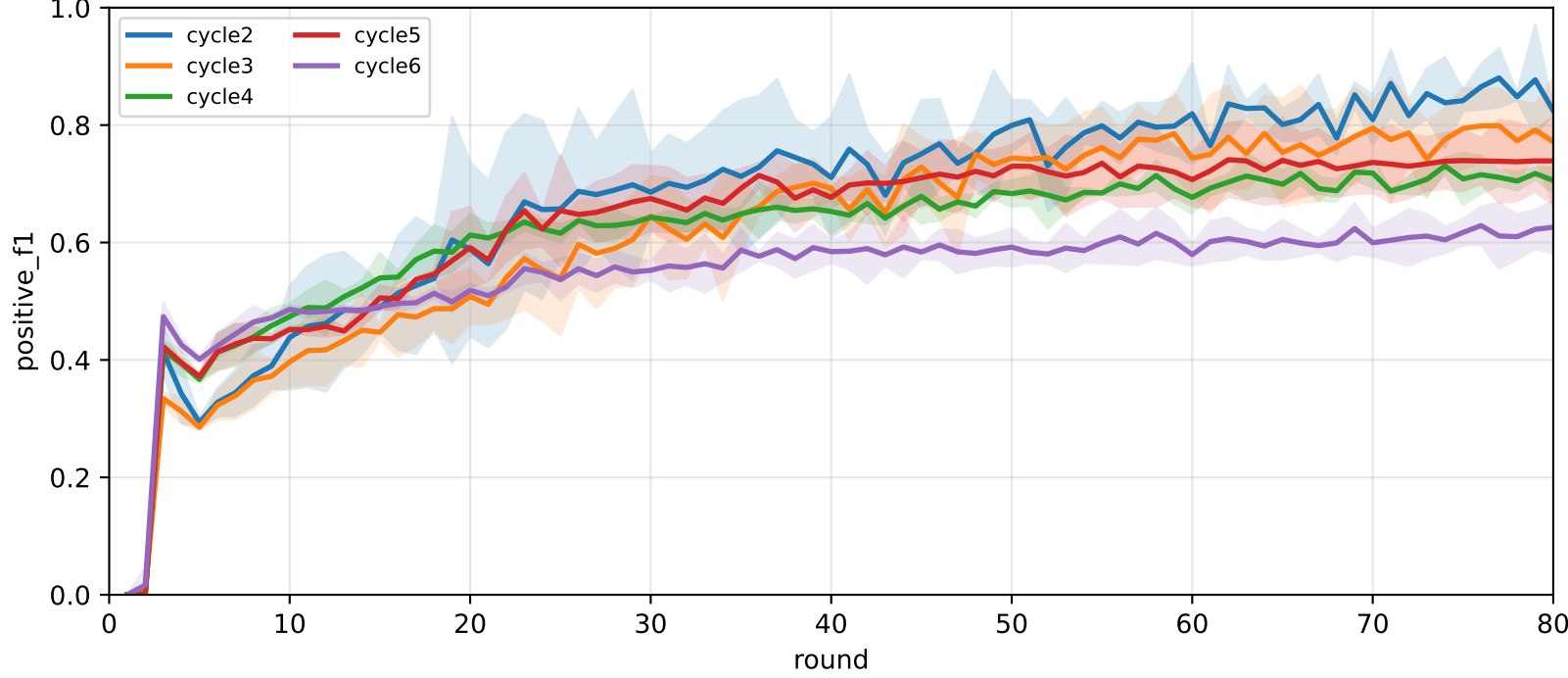
Pooled Fully Local positive F1



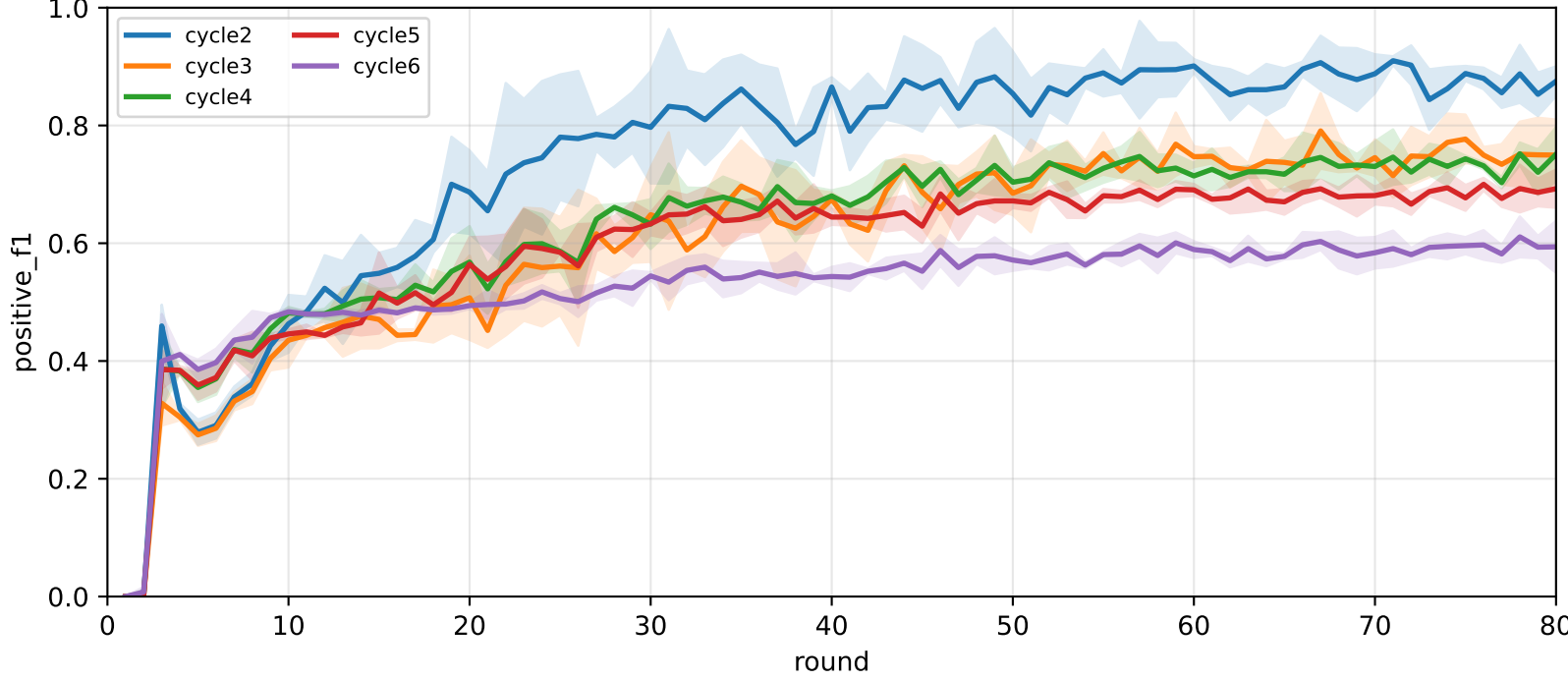
Pooled Local-Centralized positive F1



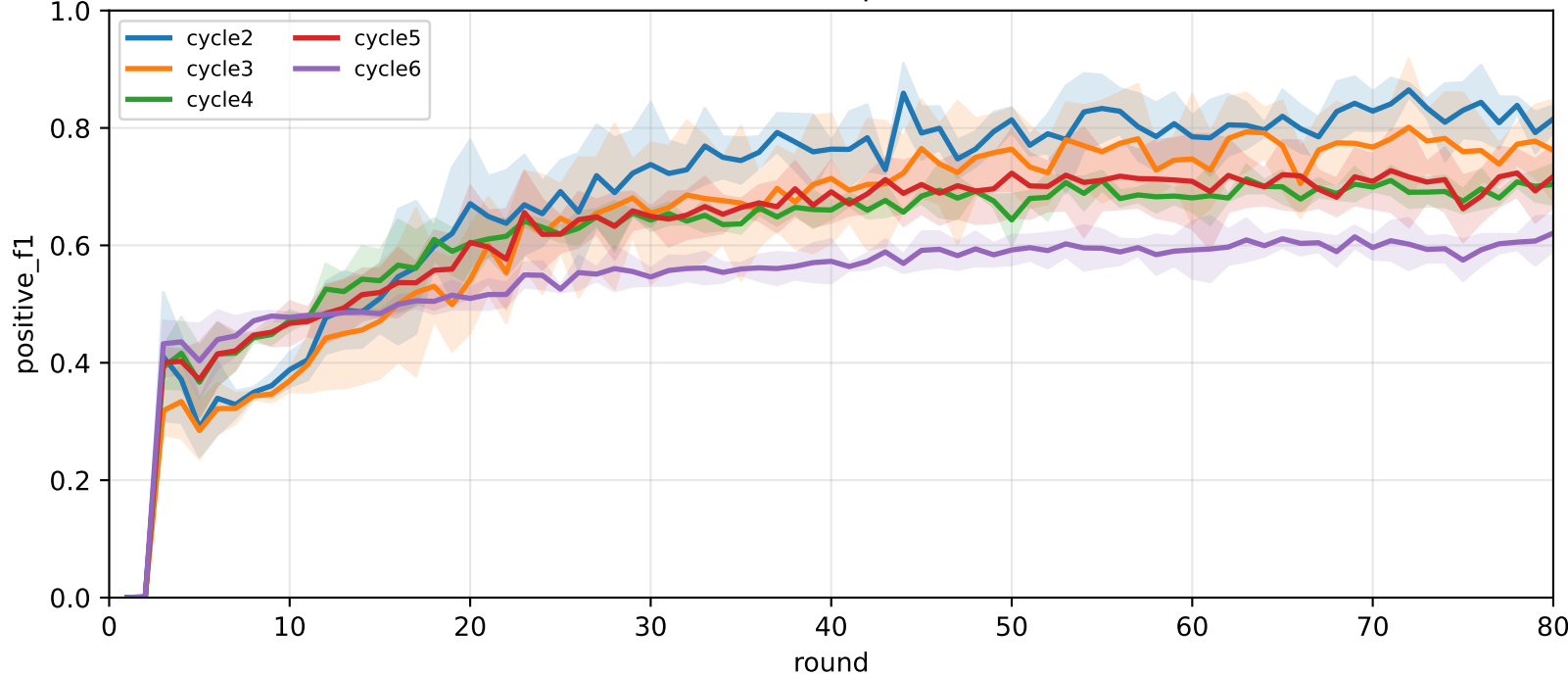
Pooled FedAvg positive F1



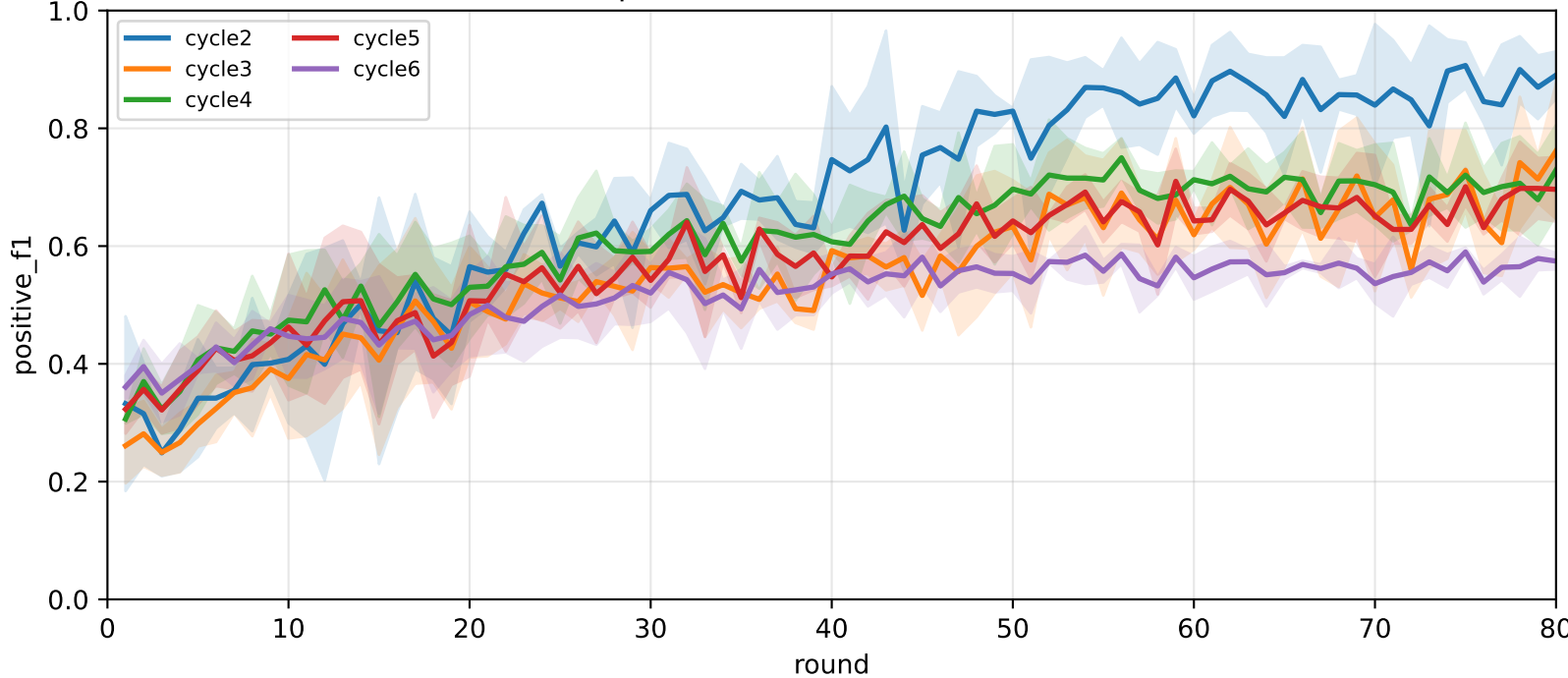
Pooled FedProx positive F1



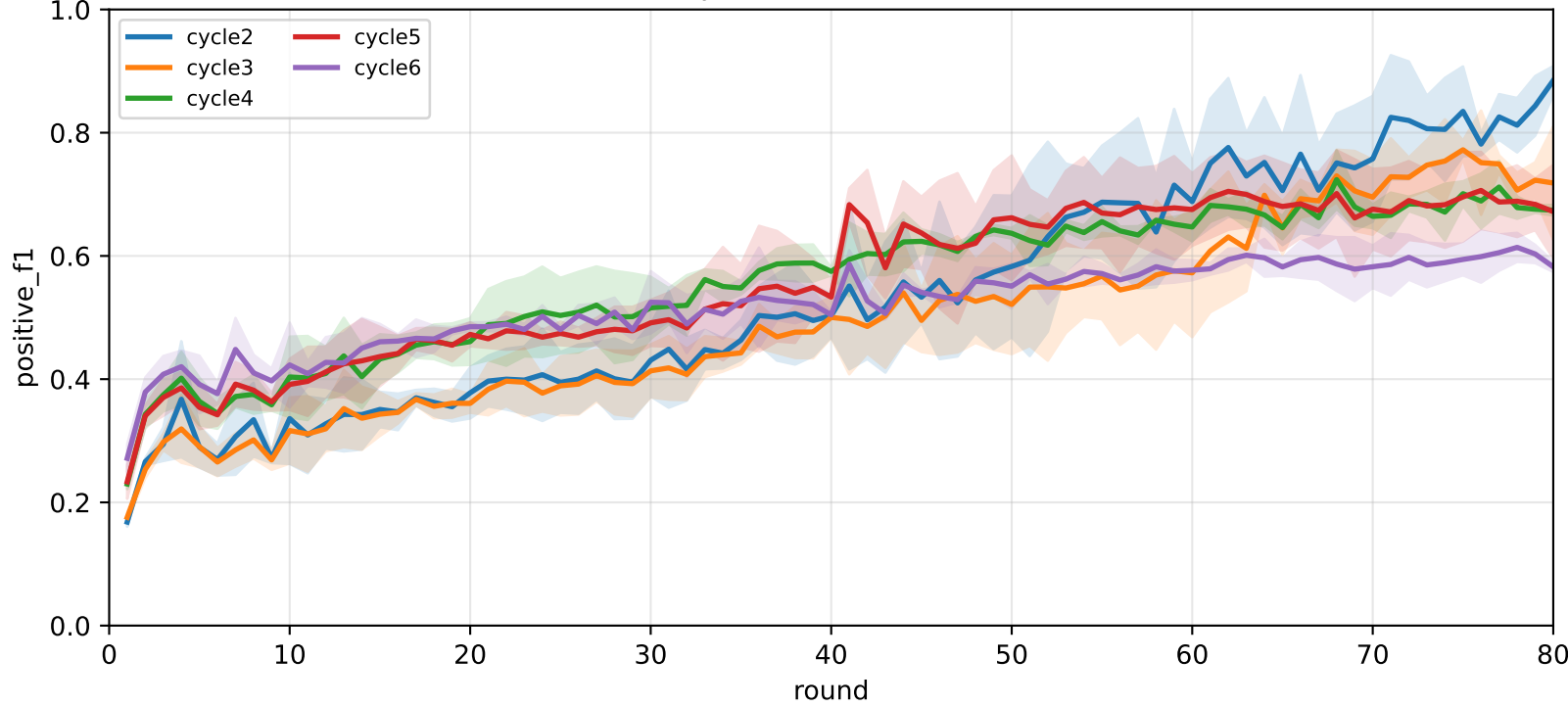
Pooled GCFL+ positive F1



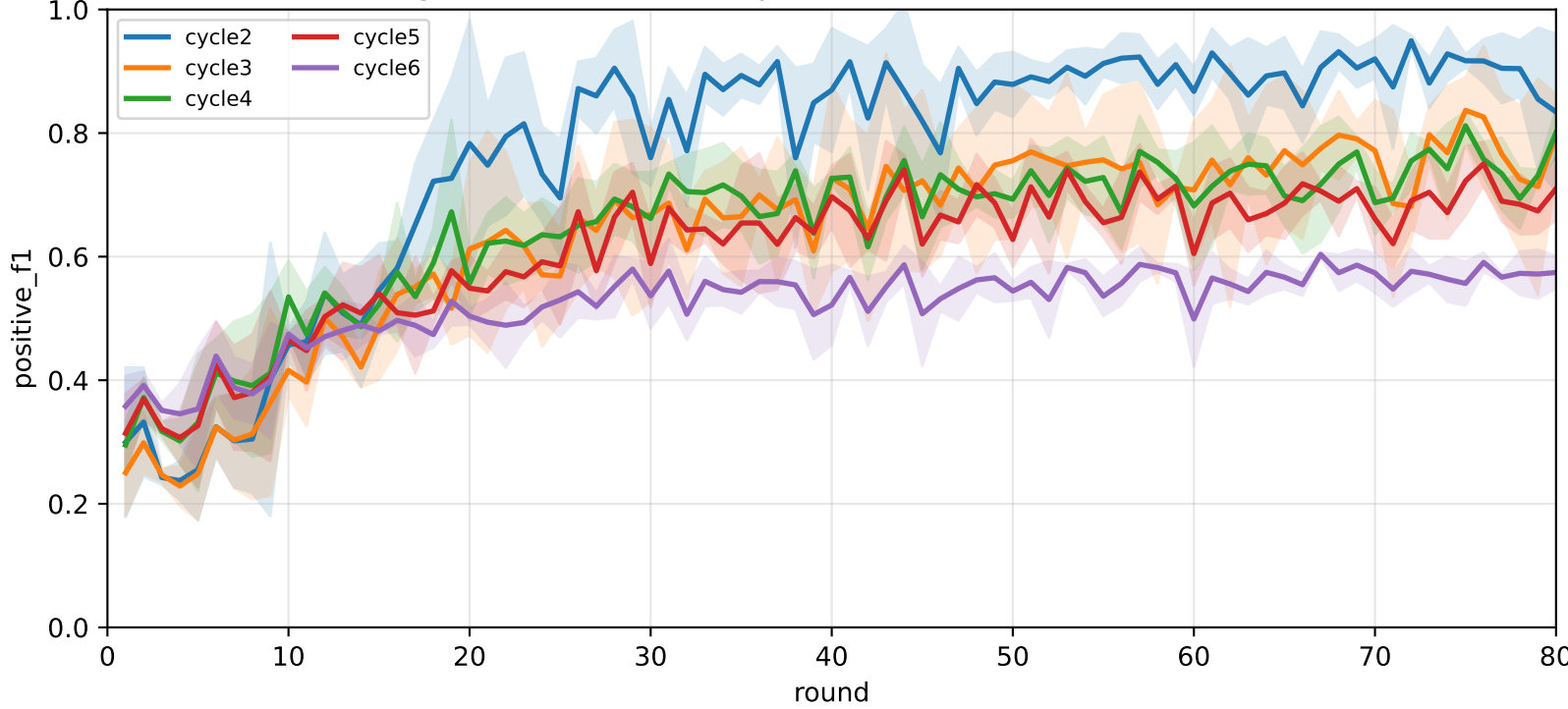
Pooled APPLE-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-Backbone-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-FedAvgBackbone-TaskHead positive F1 (after-local selected model, rounds=80)

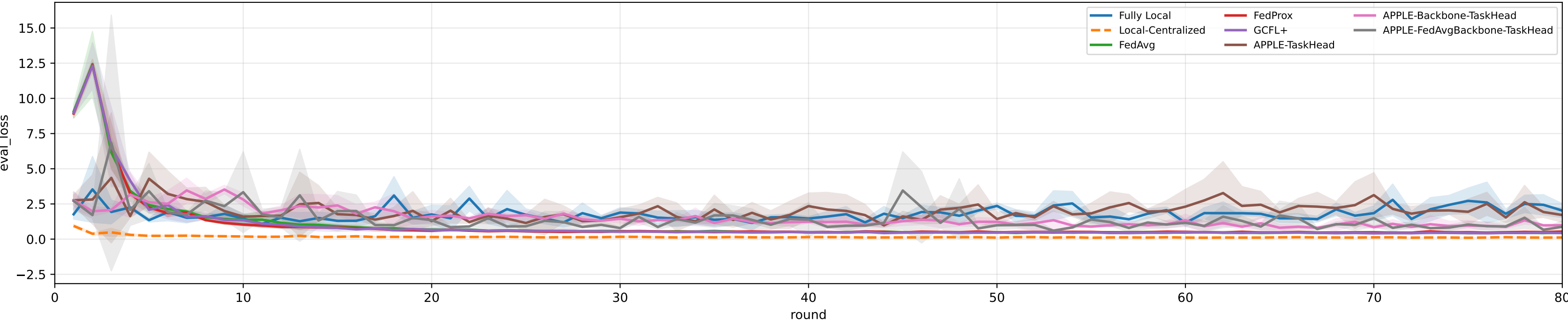


family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% |
cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

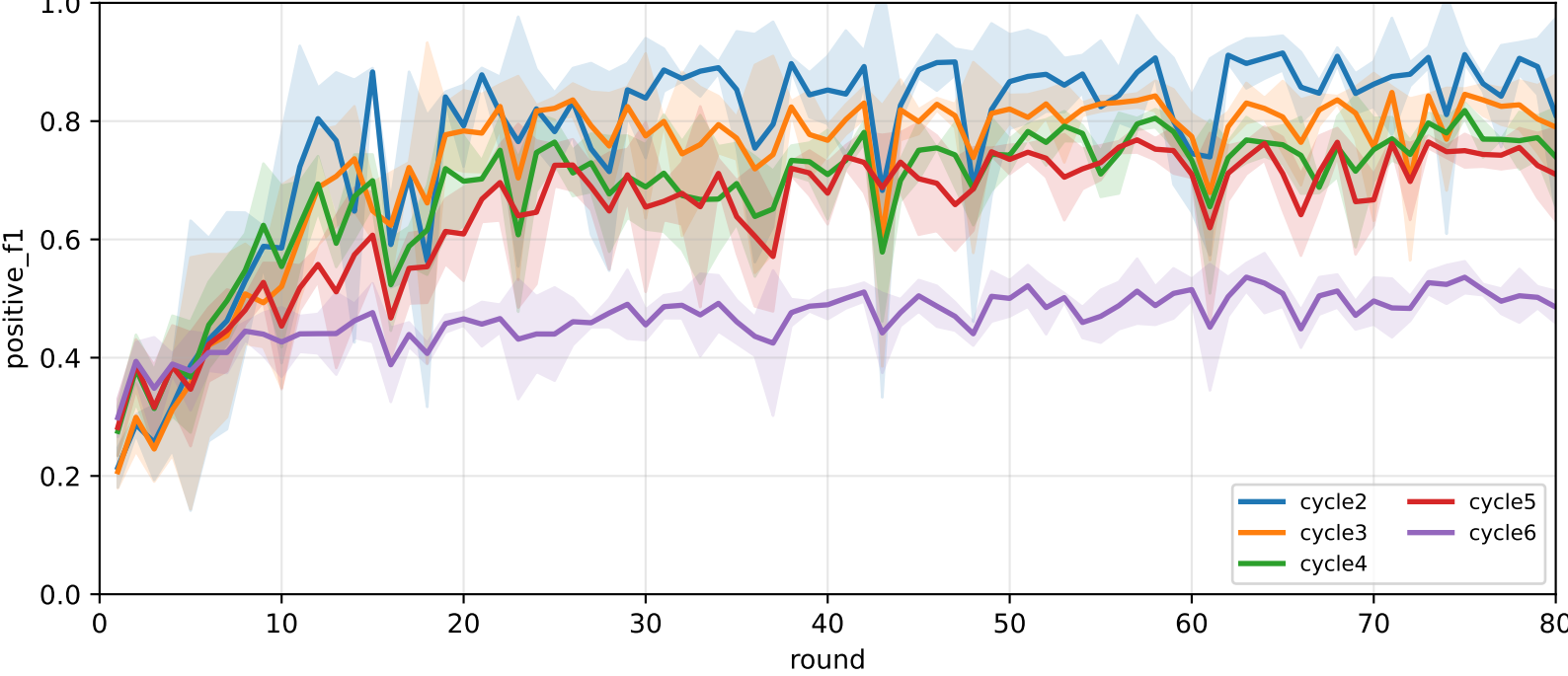
Best logged test performance (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	60.1% ± 11.6%	64.2% ± 10.9%	75.3% ± 14.5%	77.9% ± 12.0%	63.8% ± 6.1%	60.7% ± 11.5%	43.1% ± 11.1%
Local-Centralized	84.3% ± 1.6%	86.3% ± 1.4%	96.7% ± 1.1%	94.7% ± 0.6%	87.4% ± 4.2%	80.9% ± 2.2%	71.8% ± 3.9%
FedAvg	<u>70.6% ± 0.8%</u>	<u>73.2% ± 0.5%</u>	86.9% ± 4.4%	<u>82.0% ± 3.1%</u>	68.1% ± 1.9%	<u>71.0% ± 3.3%</u>	58.0% ± 1.5%
FedProx	69.0% ± 0.8%	72.0% ± 1.5%	88.7% ± 3.4%	78.9% ± 3.0%	71.2% ± 3.9%	64.6% ± 0.7%	56.4% ± 1.2%
GCFL+	69.5% ± 2.6%	71.6% ± 2.6%	82.5% ± 3.7%	81.3% ± 3.7%	67.8% ± 1.6%	68.4% ± 4.5%	<u>58.2% ± 2.4%</u>
APPLE-TaskHead	64.7% ± 3.1%	67.5% ± 3.2%	87.1% ± 7.4%	68.6% ± 5.5%	67.6% ± 6.9%	59.8% ± 6.1%	54.3% ± 3.3%
APPLE-Backbone-TaskHead	65.6% ± 2.2%	67.1% ± 2.4%	73.3% ± 7.7%	71.4% ± 5.7%	66.5% ± 1.2%	68.7% ± 5.6%	55.9% ± 2.8%
APPLE-FedAvgBackbone-TaskHead	69.3% ± 1.0%	72.3% ± 0.8%	<u>91.3% ± 2.4%</u>	72.0% ± 3.1%	<u>73.0% ± 4.5%</u>	68.6% ± 1.9%	56.4% ± 2.4%

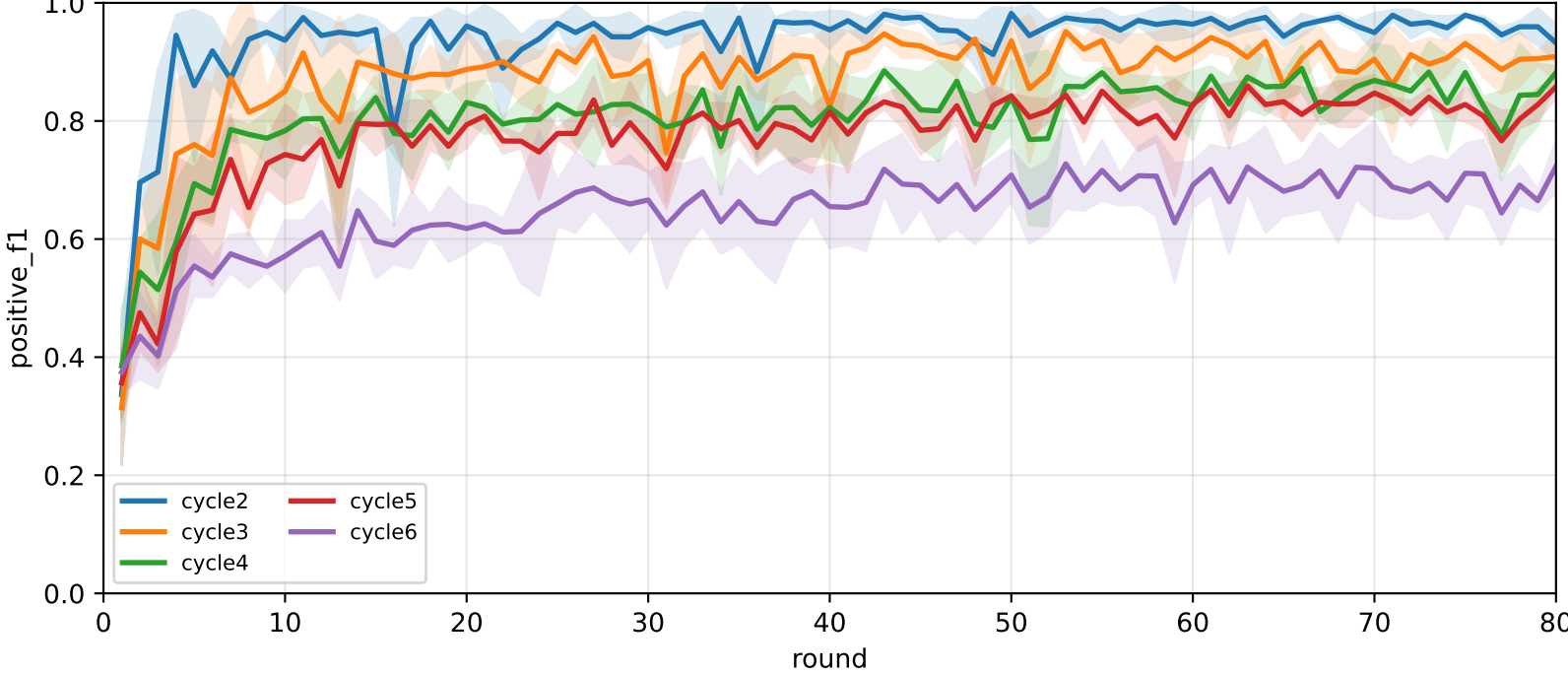
Validation loss comparison (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)



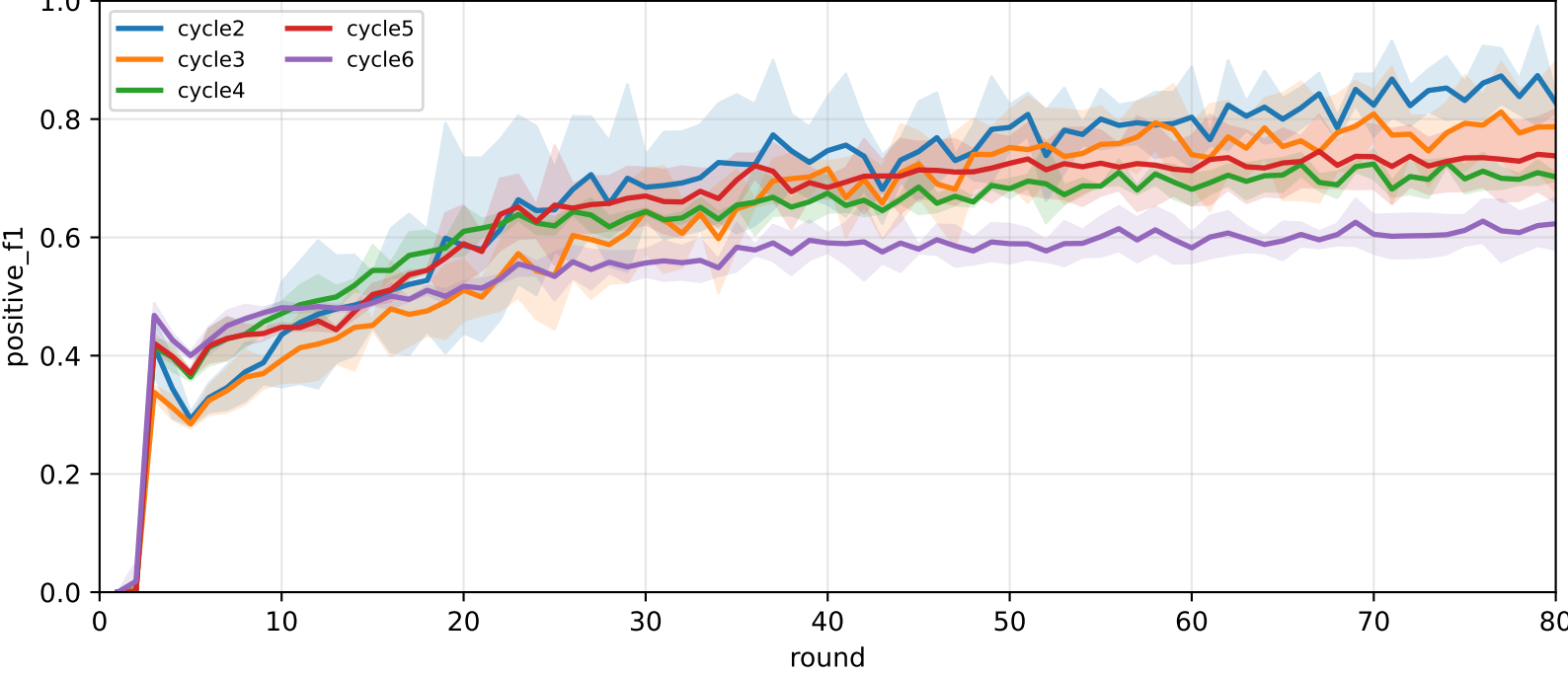
Pooled Fully Local positive F1



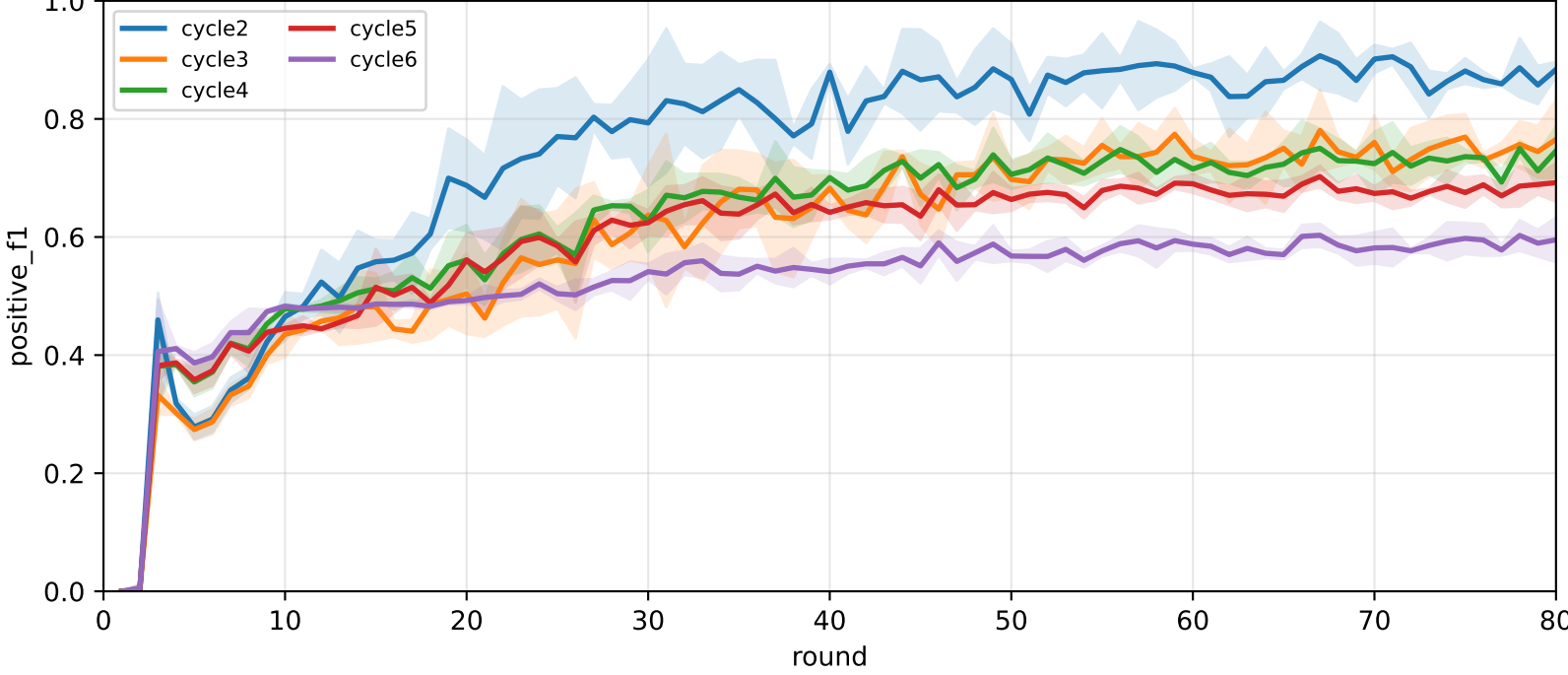
Pooled Local-Centralized positive F1



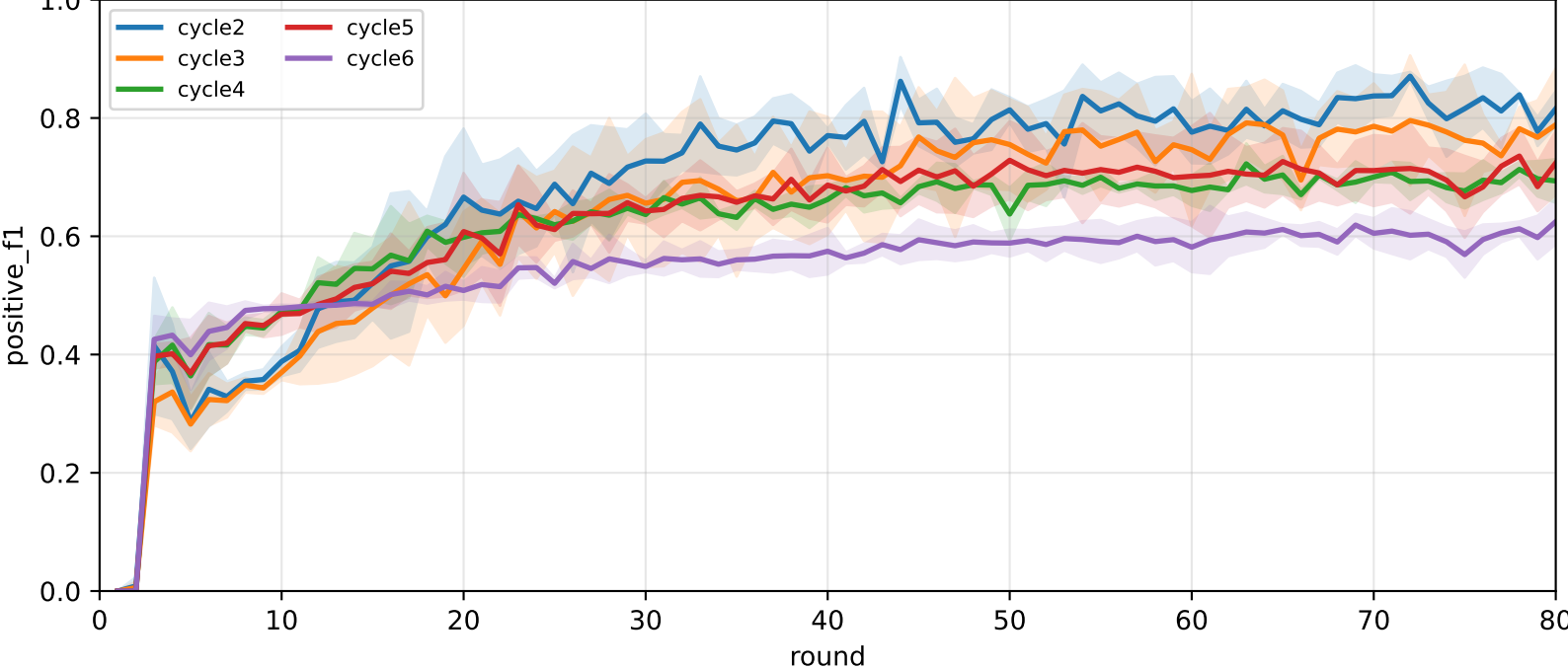
Pooled FedAvg positive F1



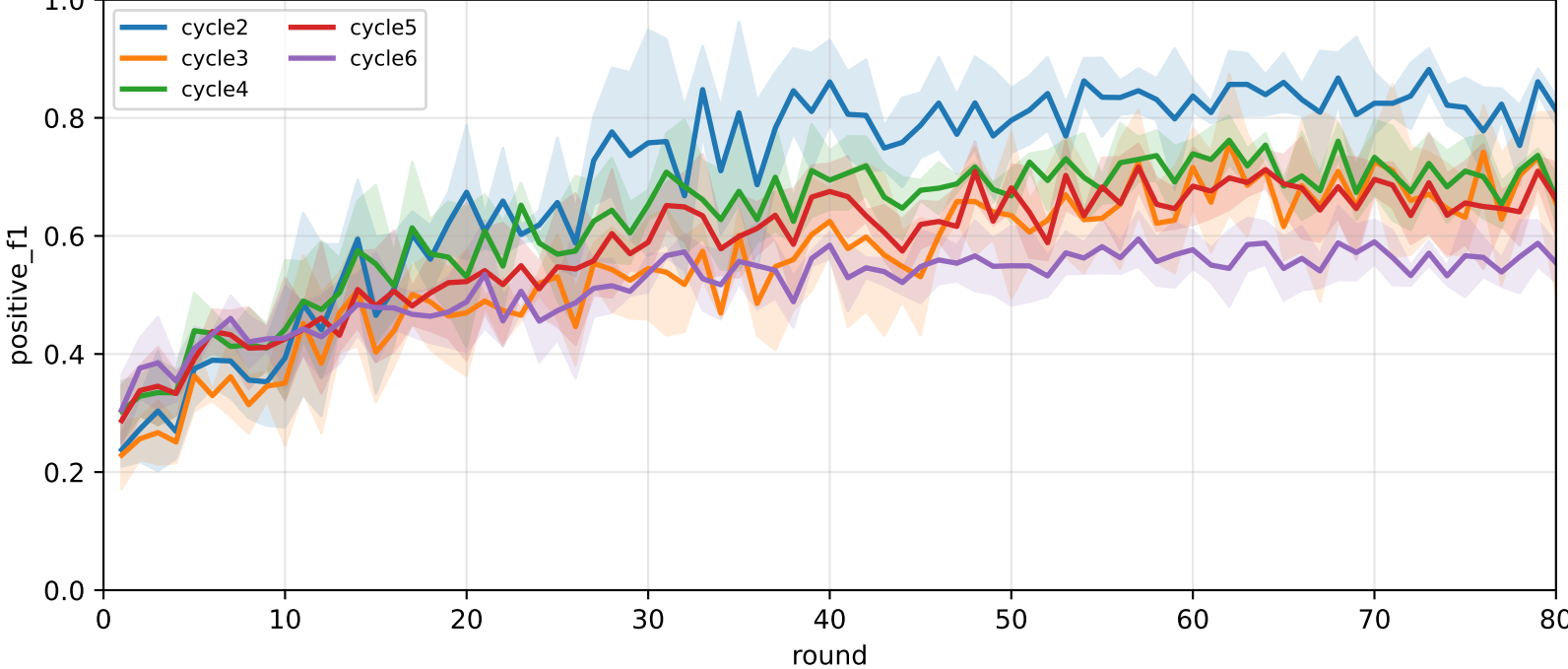
Pooled FedProx positive F1



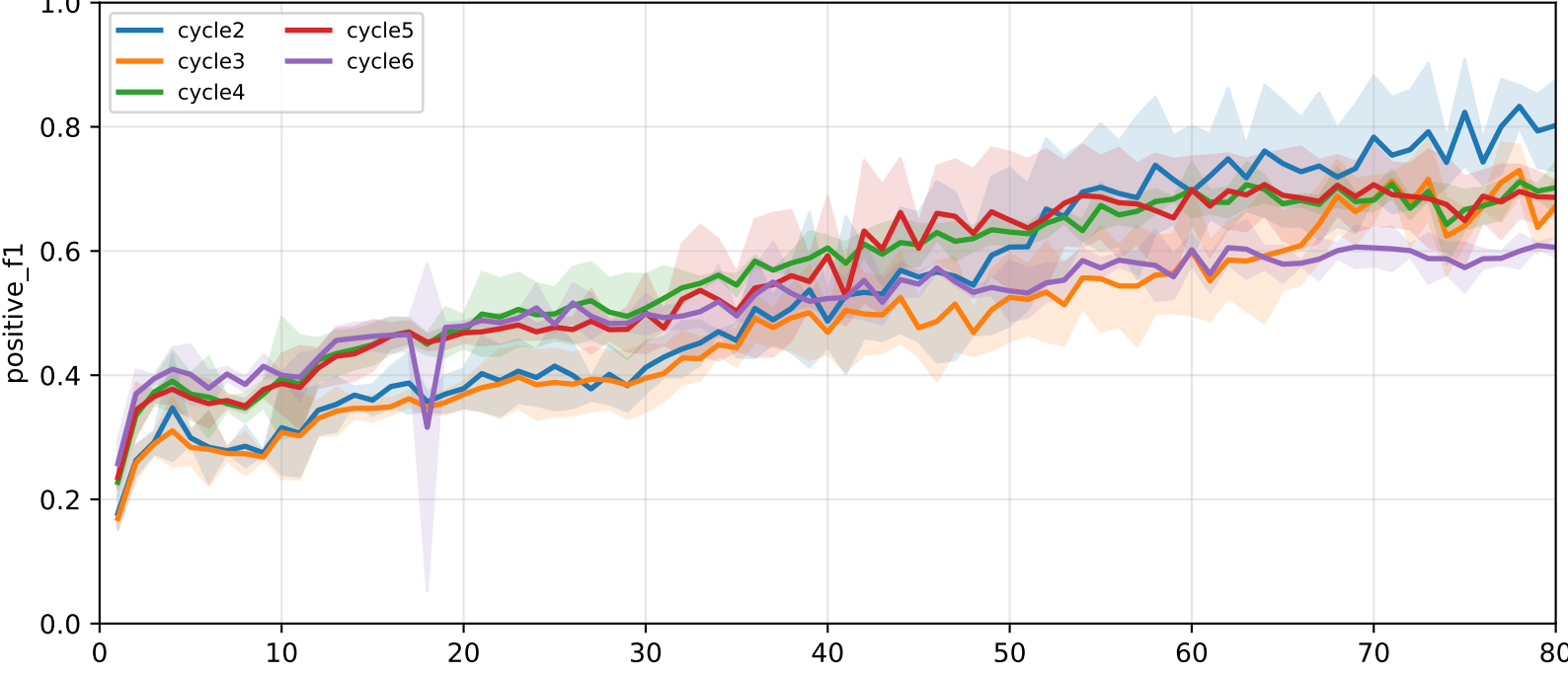
Pooled GCFL+ positive F1



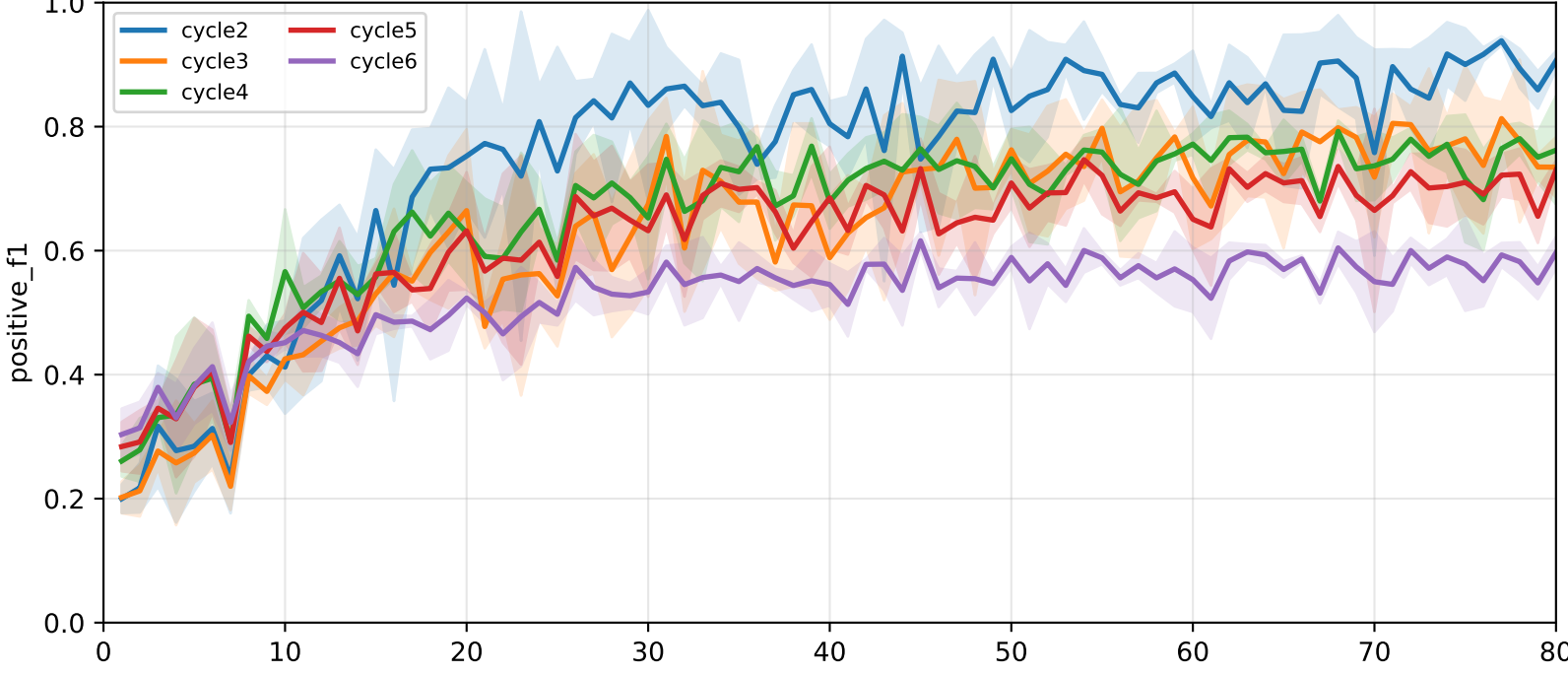
Pooled APPLE-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-Backbone-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-FedAvgBackbone-TaskHead positive F1 (after-local selected model, rounds=80)



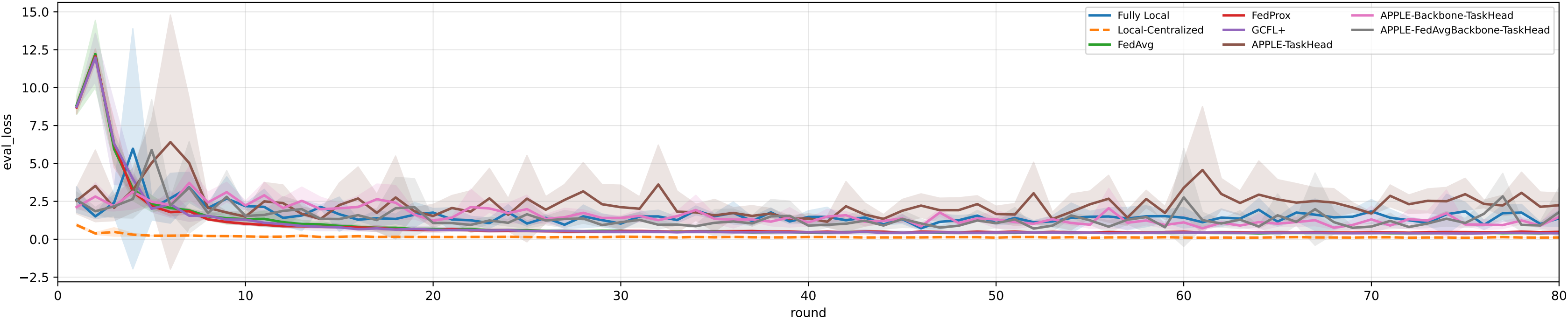
Controlled q-label heterogeneity: Mid | family aggregated / pooled performance |
family=q_task_cycle4 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% |
cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

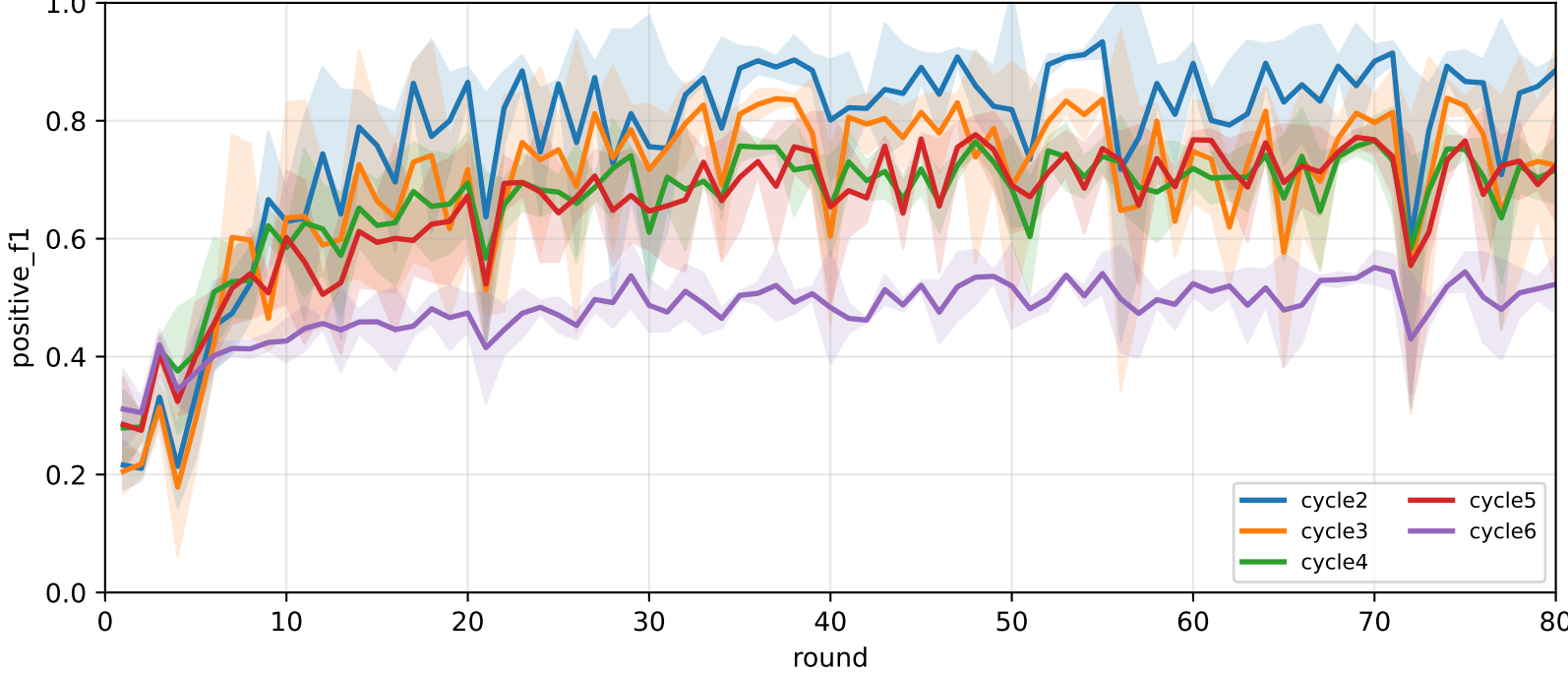
Best logged test performance (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	58.7% ± 5.5%	63.8% ± 5.1%	82.7% ± 2.7%	76.5% ± 5.8%	63.3% ± 5.5%	54.2% ± 7.8%	42.1% ± 3.8%
Local-Centralized	84.3% ± 1.6%	86.3% ± 1.4%	96.7% ± 1.1%	94.7% ± 0.6%	87.4% ± 4.2%	80.9% ± 2.2%	71.8% ± 3.9%
FedAvg	<u>69.9% ± 1.5%</u>	<u>72.5% ± 1.1%</u>	86.4% ± 4.4%	<u>81.6% ± 4.2%</u>	67.7% ± 1.9%	<u>69.0% ± 4.4%</u>	57.9% ± 1.9%
FedProx	68.8% ± 0.5%	72.0% ± 1.1%	<u>89.8% ± 3.6%</u>	79.1% ± 2.5%	<u>71.3% ± 2.9%</u>	63.6% ± 1.7%	56.1% ± 1.2%
GCFL+	69.4% ± 2.8%	71.7% ± 2.8%	83.3% ± 2.6%	81.4% ± 4.0%	67.6% ± 1.6%	68.1% ± 4.3%	<u>58.0% ± 2.5%</u>
APPLE-TaskHead	57.1% ± 3.7%	58.8% ± 3.3%	69.4% ± 7.5%	62.3% ± 2.3%	54.7% ± 5.0%	57.9% ± 7.0%	49.6% ± 4.5%
APPLE-Backbone-TaskHead	62.9% ± 3.3%	63.9% ± 3.5%	65.9% ± 4.9%	71.9% ± 5.7%	62.1% ± 2.3%	64.4% ± 6.3%	55.2% ± 2.9%
APPLE-FedAvgBackbone-TaskHead	66.7% ± 5.5%	69.7% ± 5.4%	84.2% ± 5.3%	76.1% ± 4.6%	69.7% ± 6.9%	66.1% ± 6.2%	52.3% ± 5.4%

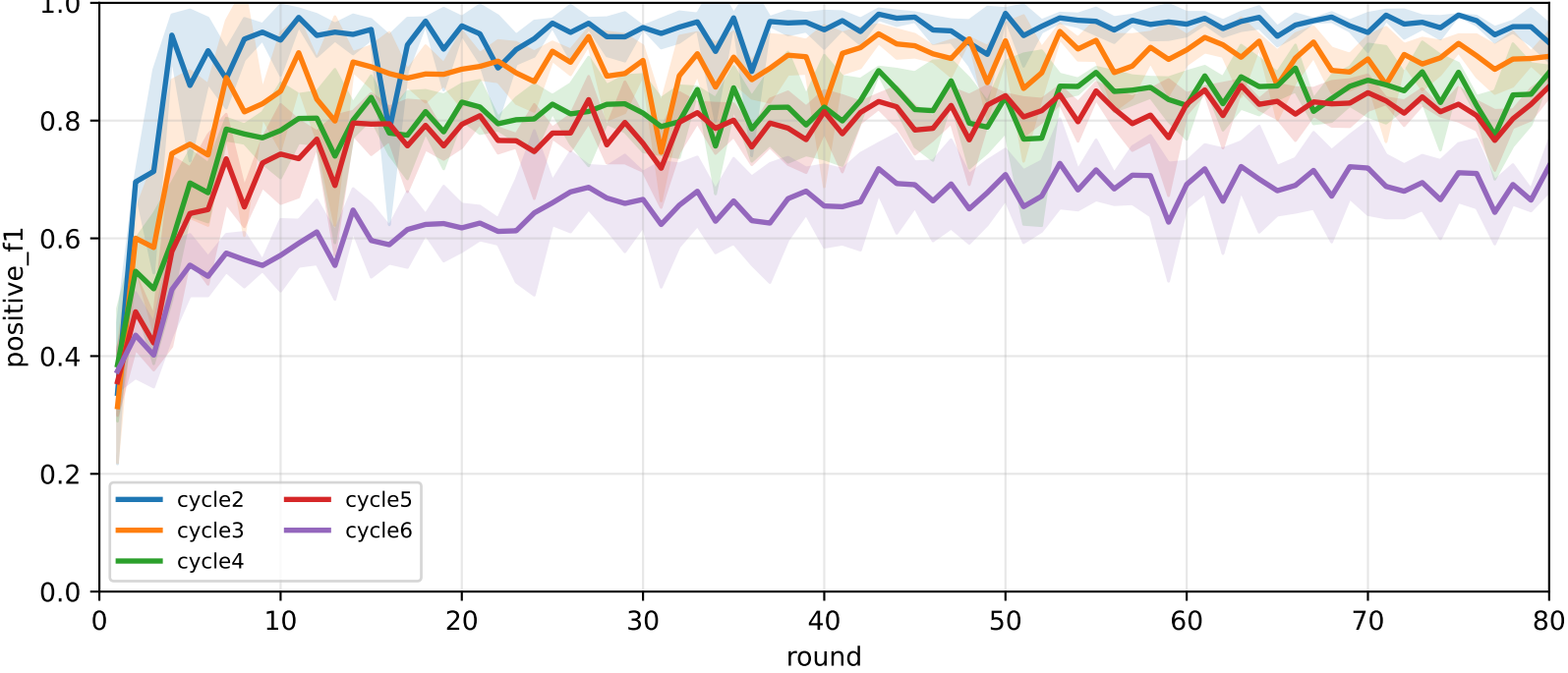
Validation loss comparison (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)



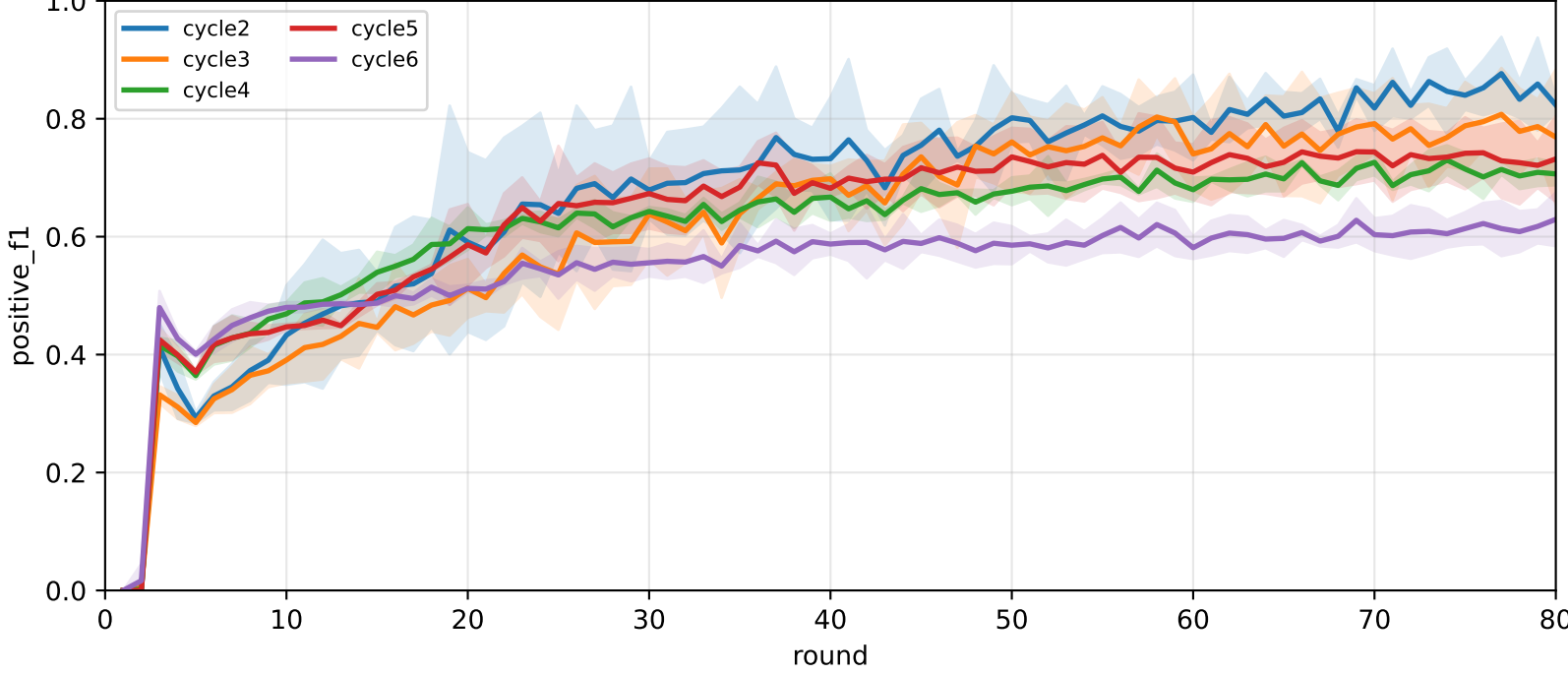
Pooled Fully Local positive F1



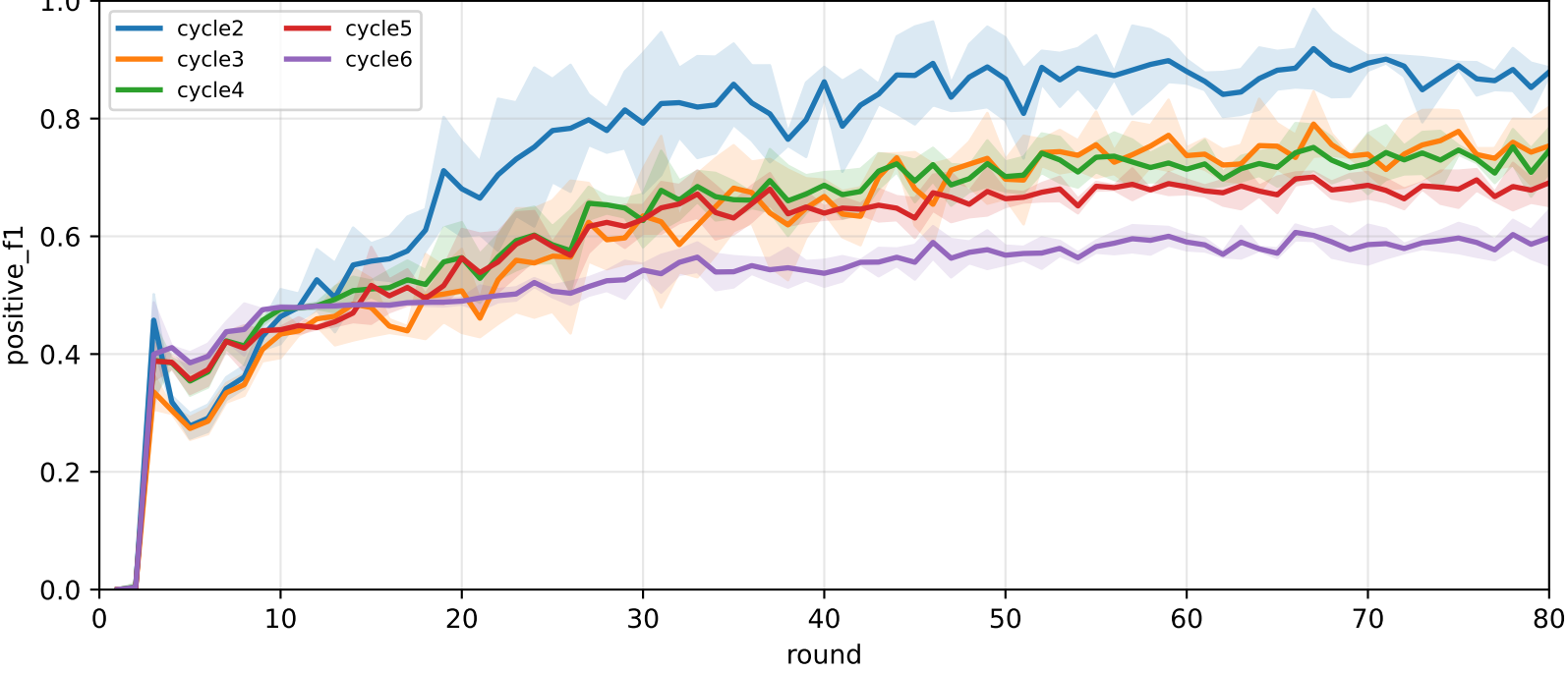
Pooled Local-Centralized positive F1



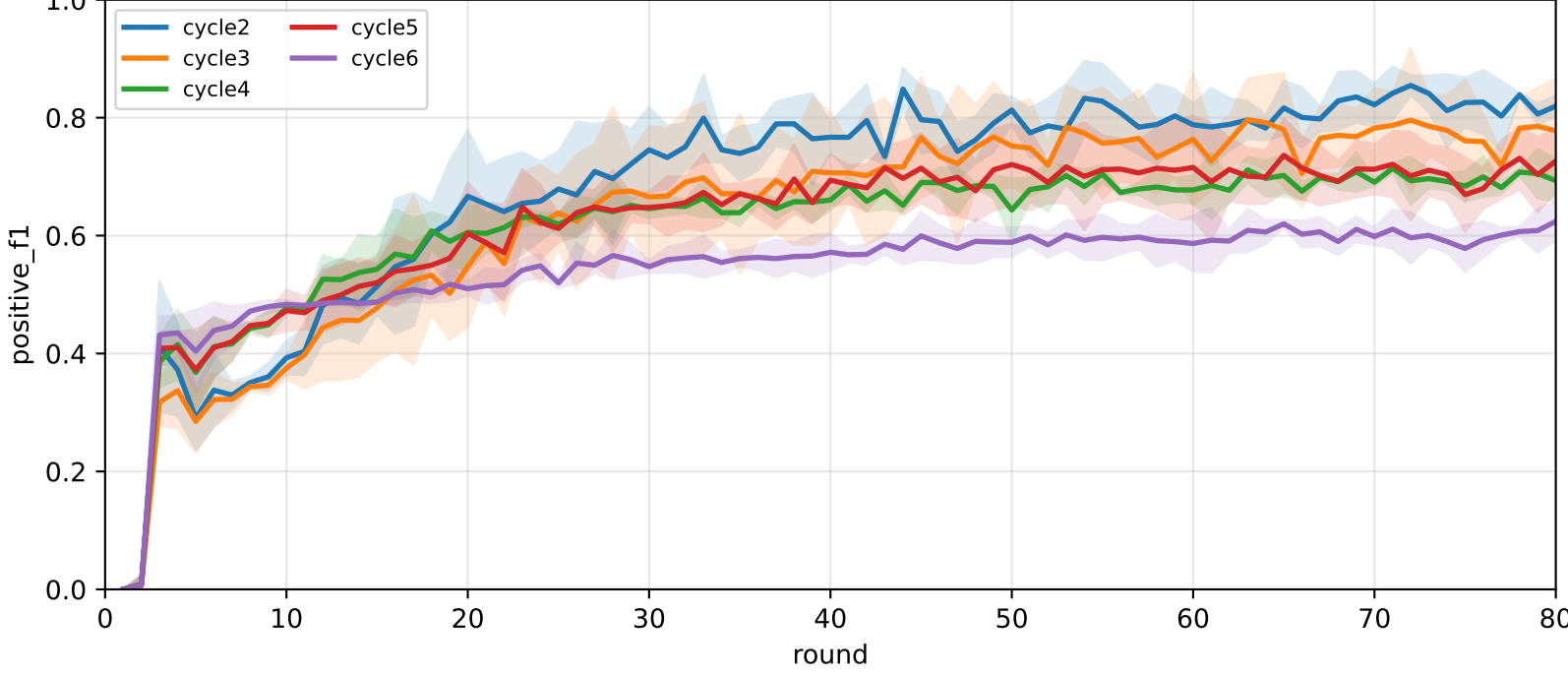
Pooled FedAvg positive F1



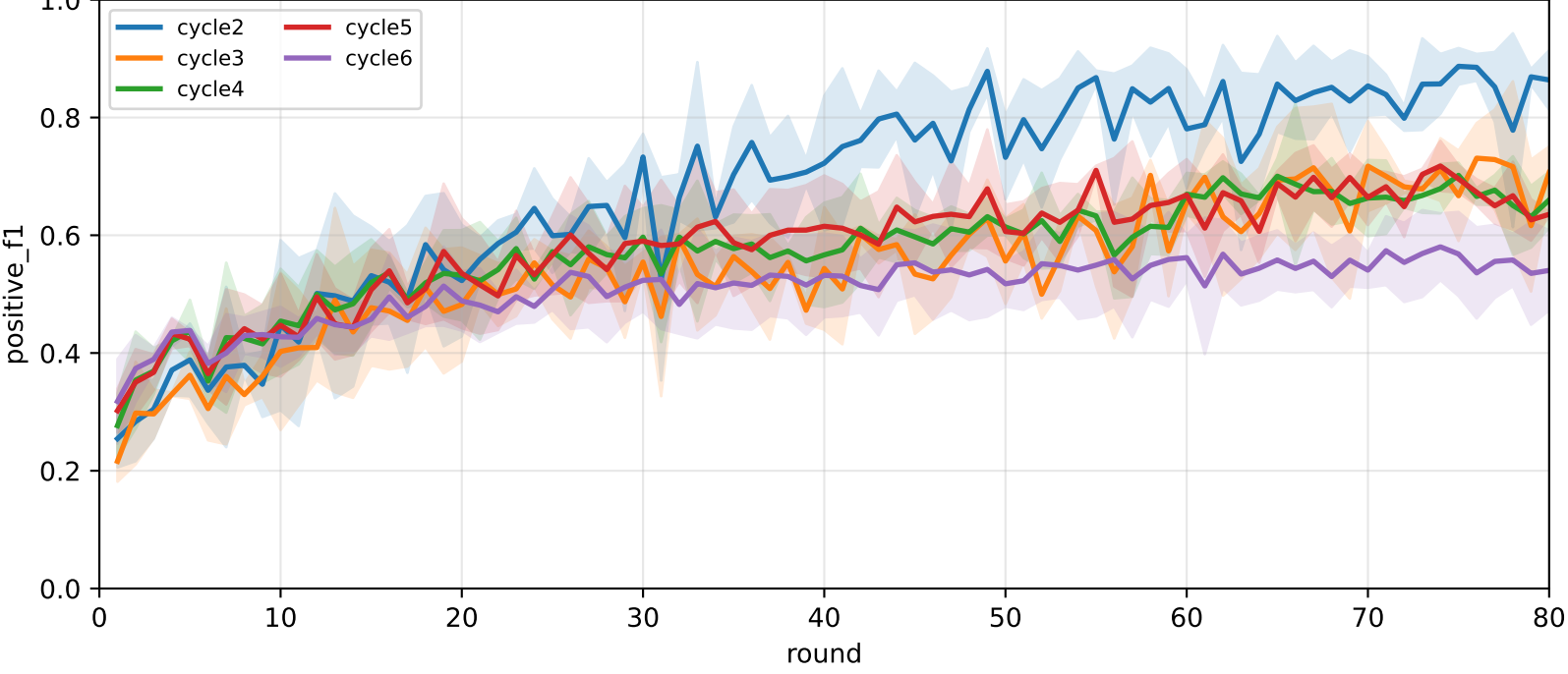
Pooled FedProx positive F1



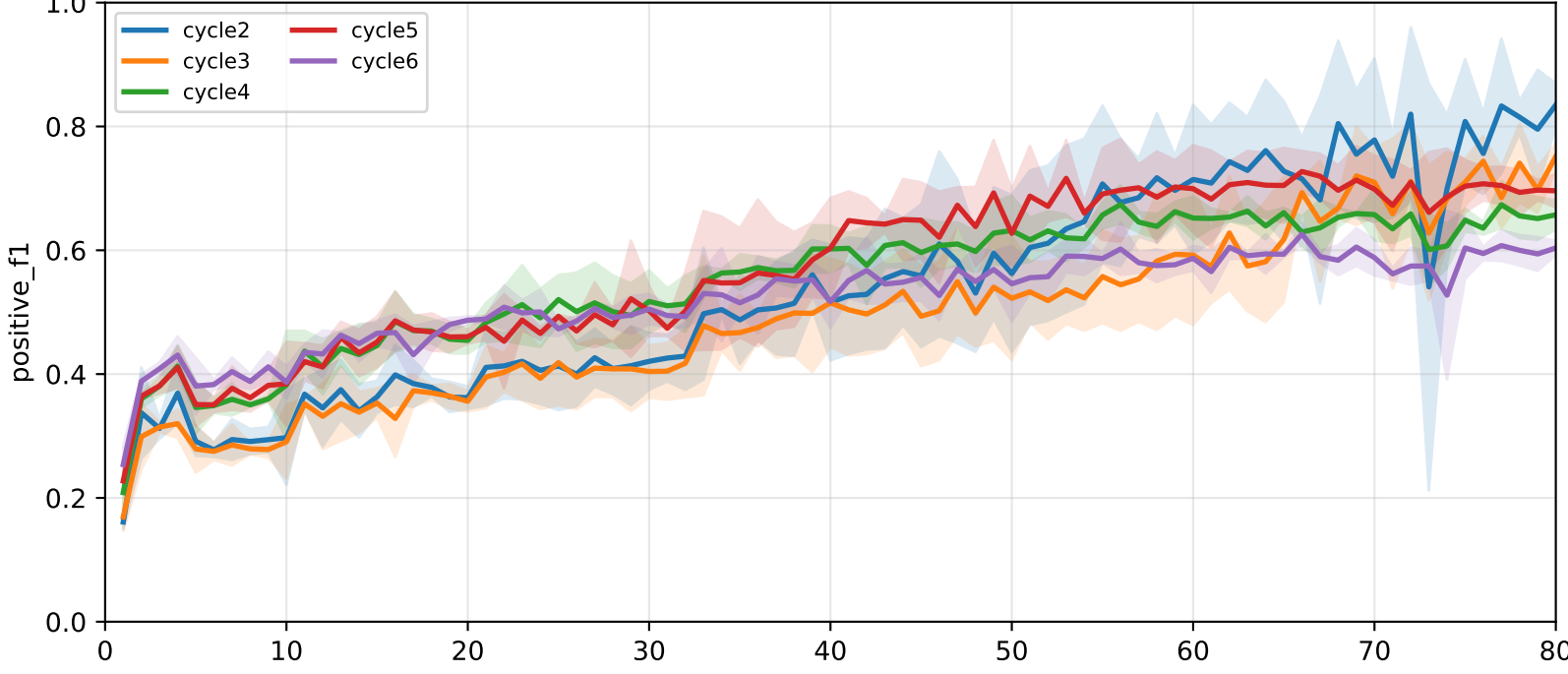
Pooled GCFL+ positive F1



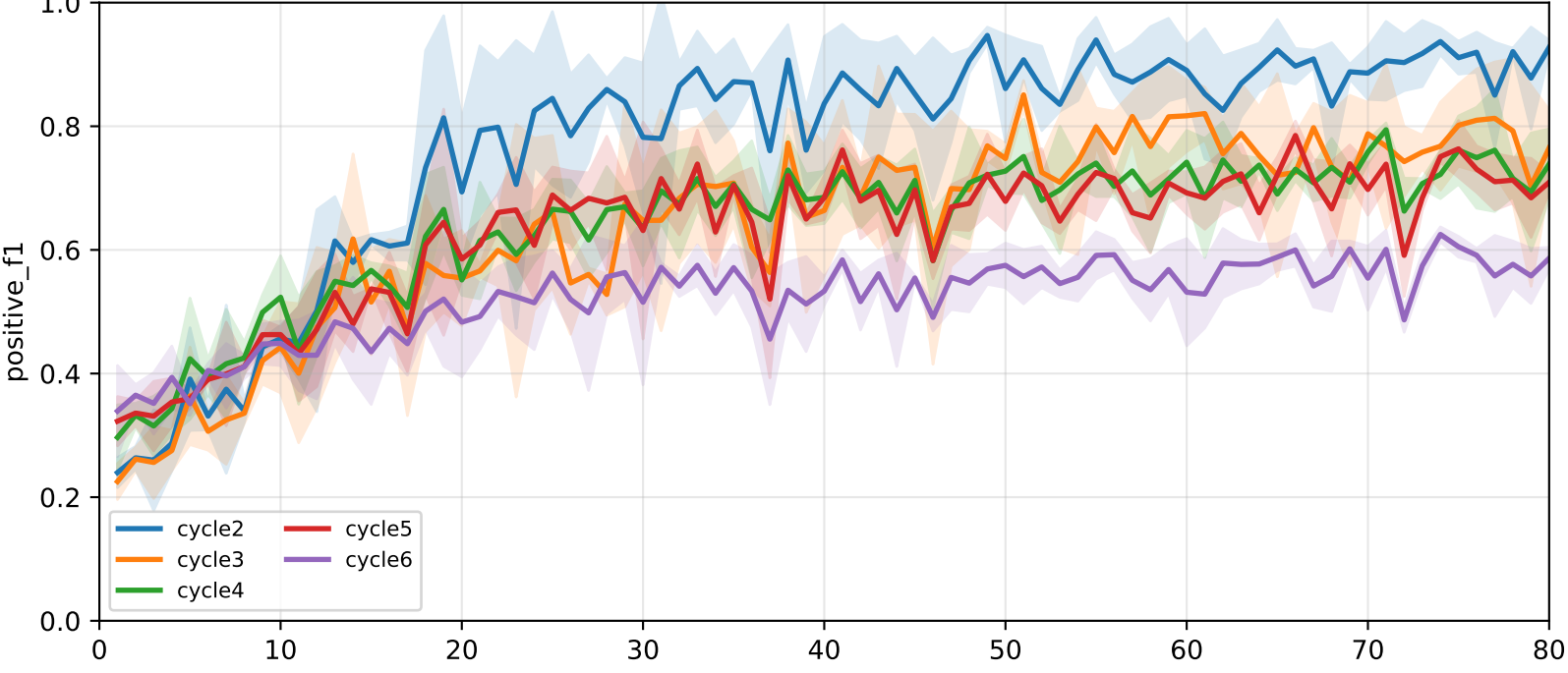
Pooled APPLE-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-Backbone-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-FedAvgBackbone-TaskHead positive F1 (after-local selected model, rounds=80)



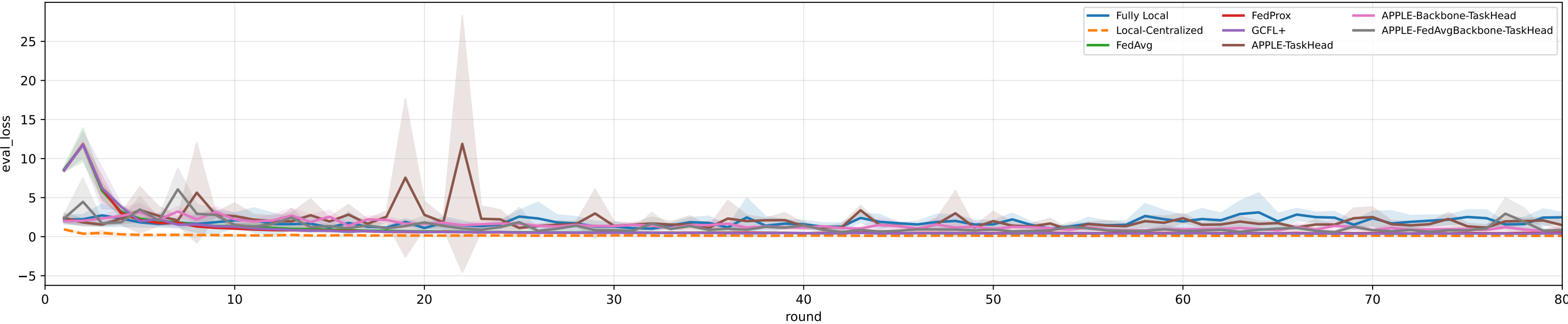
Controlled q-label heterogeneity: Mid | family aggregated / pooled performance |
family=q task_cycle5 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% |
cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

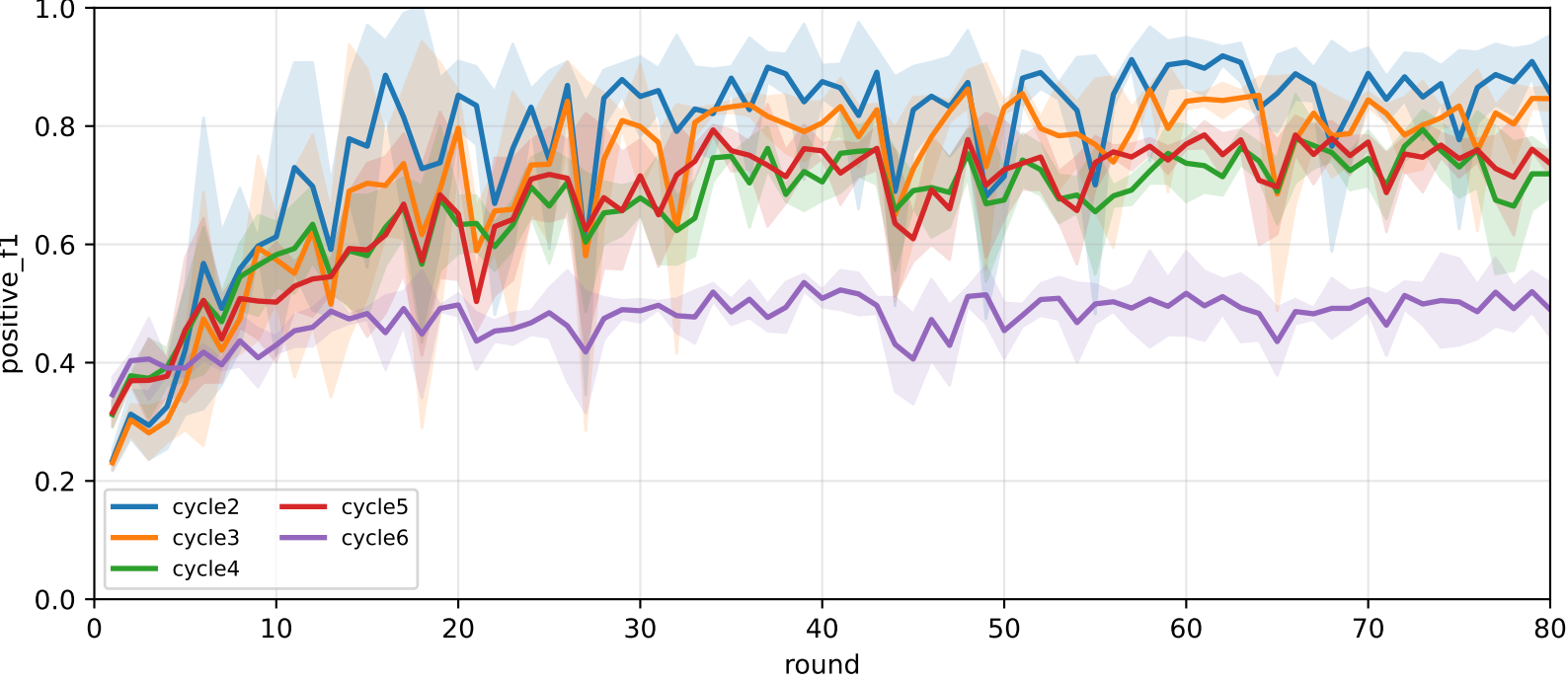
Best logged test performance (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	61.2% ± 0.9%	66.3% ± 1.4%	86.3% ± 4.7%	79.8% ± 5.8%	63.6% ± 1.5%	58.1% ± 1.4%	43.8% ± 3.4%
Local-Centralized	84.3% ± 1.6%	86.3% ± 1.4%	96.7% ± 1.1%	94.7% ± 0.6%	87.4% ± 4.2%	80.9% ± 2.2%	71.8% ± 3.9%
FedAvg	70.5% ± 1.4%	73.0% ± 1.2%	85.8% ± 4.7%	82.2% ± 4.3%	68.2% ± 2.1%	<u>70.4% ± 4.3%</u>	58.5% ± 2.0%
FedProx	69.0% ± 0.9%	72.0% ± 1.4%	88.4% ± 3.4%	79.4% ± 2.4%	71.4% ± 4.2%	64.3% ± 1.5%	56.5% ± 1.2%
GCFL+	69.4% ± 2.6%	71.7% ± 2.5%	83.2% ± 3.9%	82.2% ± 3.7%	68.0% ± 1.5%	67.4% ± 3.8%	57.8% ± 2.8%
APPLE-TaskHead	58.5% ± 2.8%	59.5% ± 2.9%	66.7% ± 6.7%	57.9% ± 3.7%	65.3% ± 1.6%	54.8% ± 3.1%	53.1% ± 3.6%
APPLE-Backbone-TaskHead	65.0% ± 1.8%	66.4% ± 2.5%	71.8% ± 9.0%	74.6% ± 5.3%	64.9% ± 0.6%	64.7% ± 4.5%	56.1% ± 1.8%
APPLE-FedAvgBackbone-TaskHead	<u>73.0% ± 0.8%</u>	<u>75.6% ± 1.4%</u>	<u>89.5% ± 5.1%</u>	<u>84.7% ± 5.5%</u>	<u>76.1% ± 3.9%</u>	67.8% ± 2.9%	<u>60.2% ± 0.5%</u>

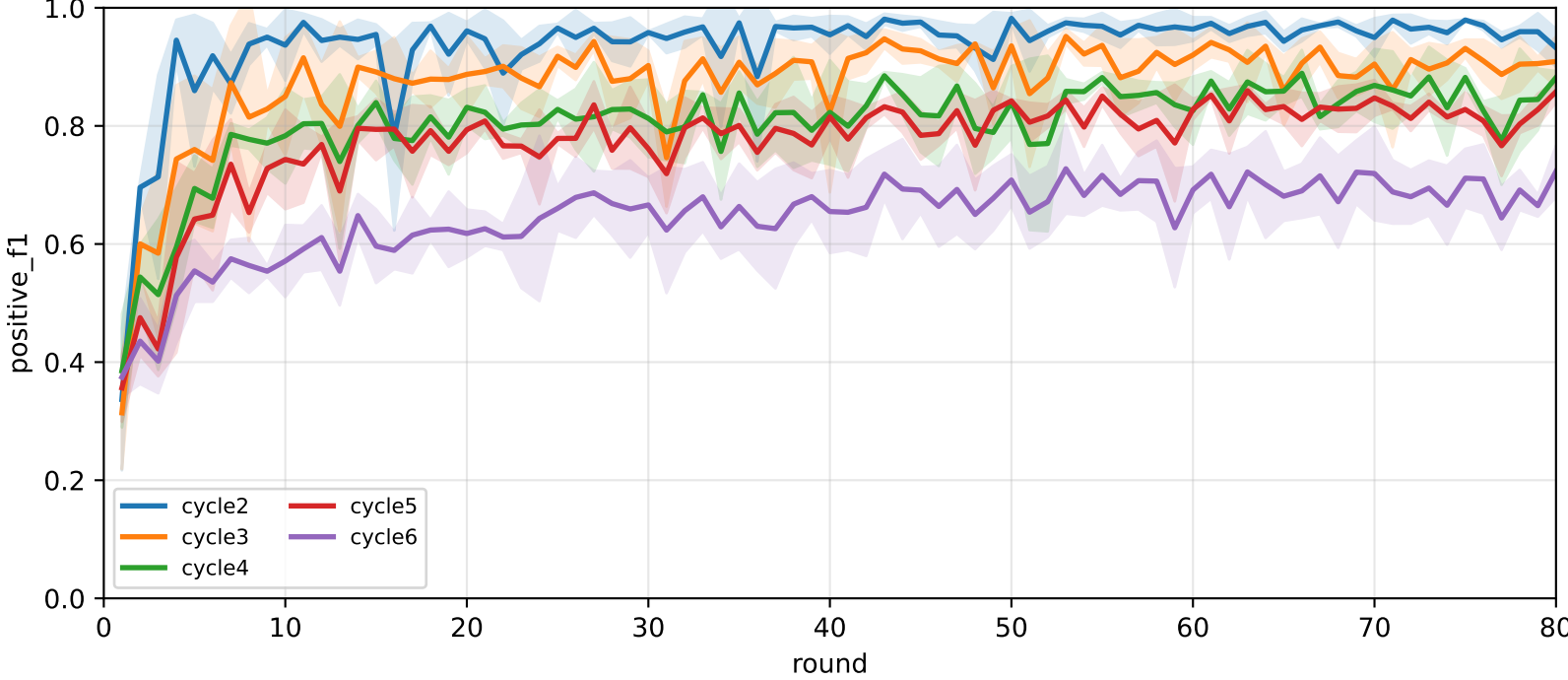
Validation loss comparison (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)



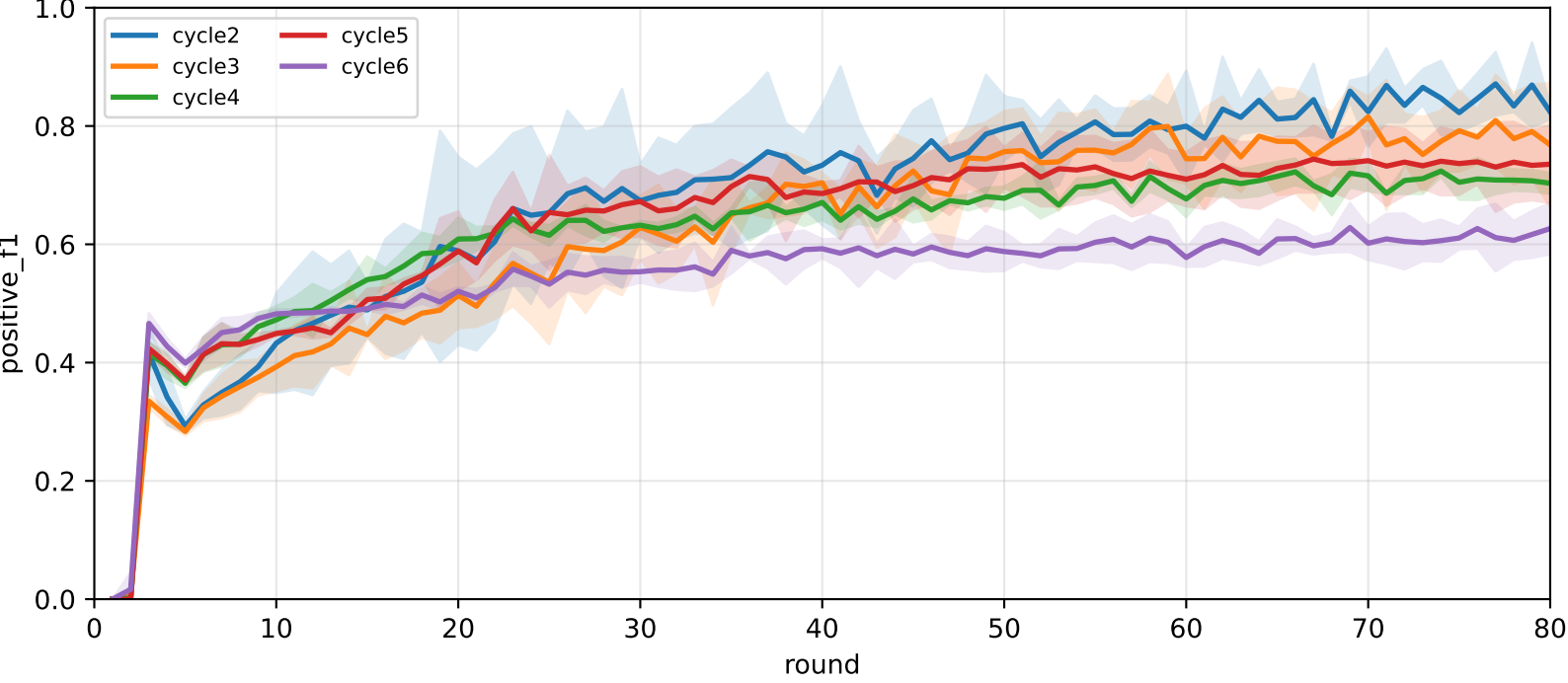
Pooled Fully Local positive F1



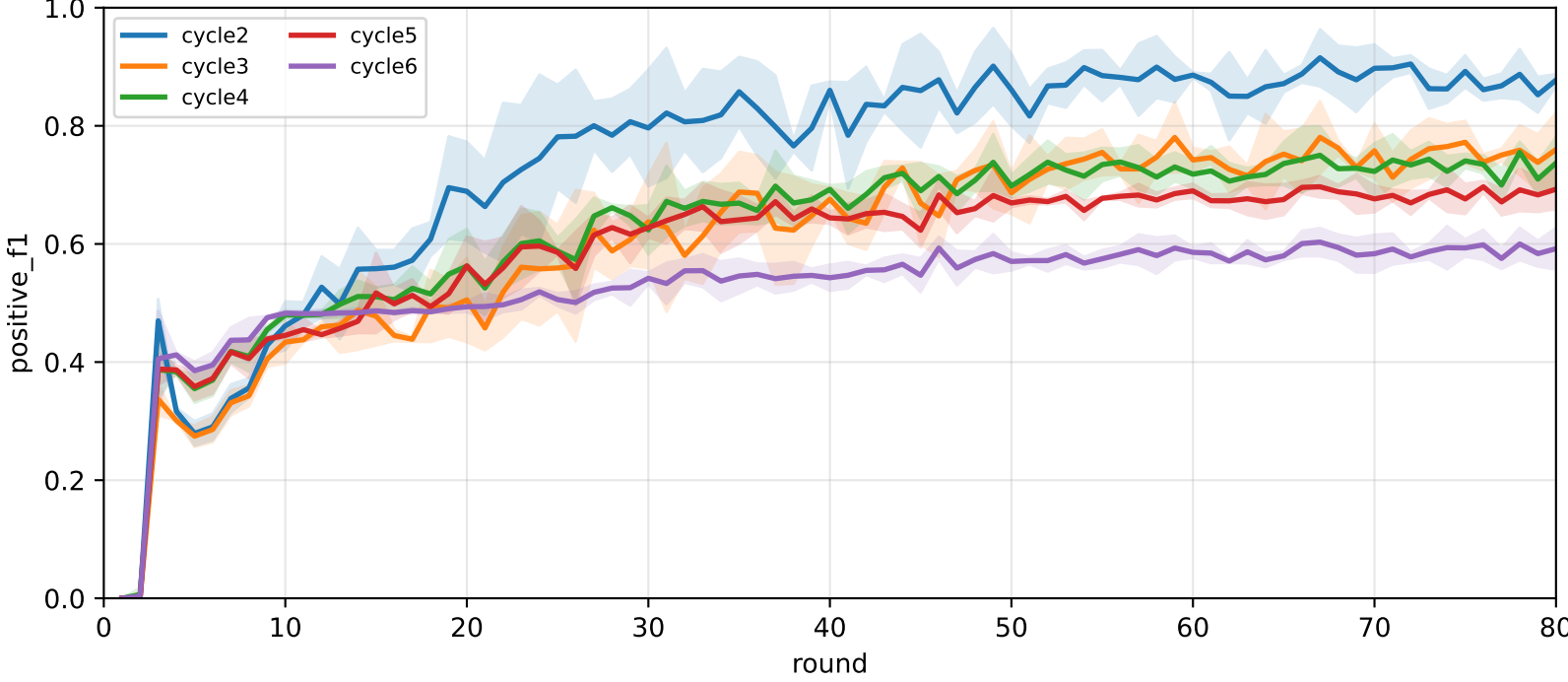
Pooled Local-Centralized positive F1



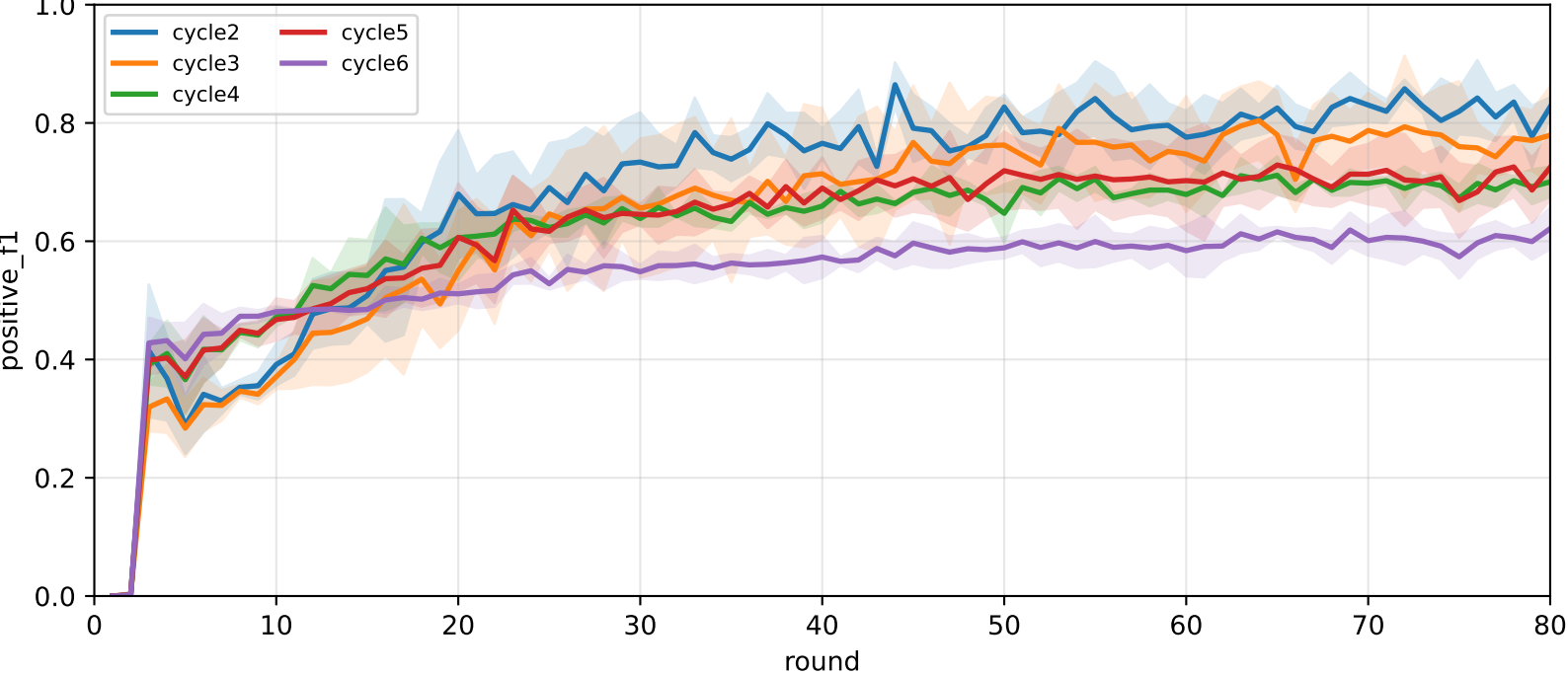
Pooled FedAvg positive F1



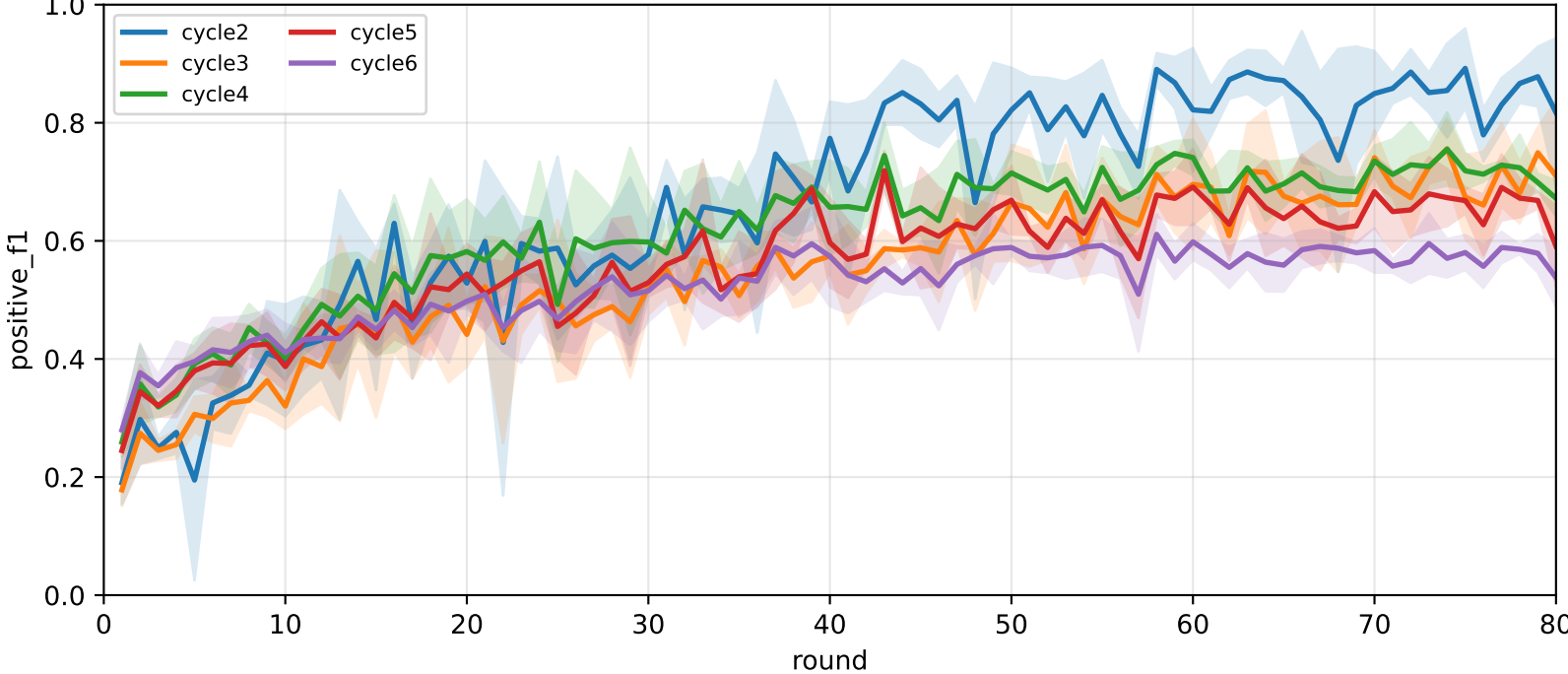
Pooled FedProx positive F1



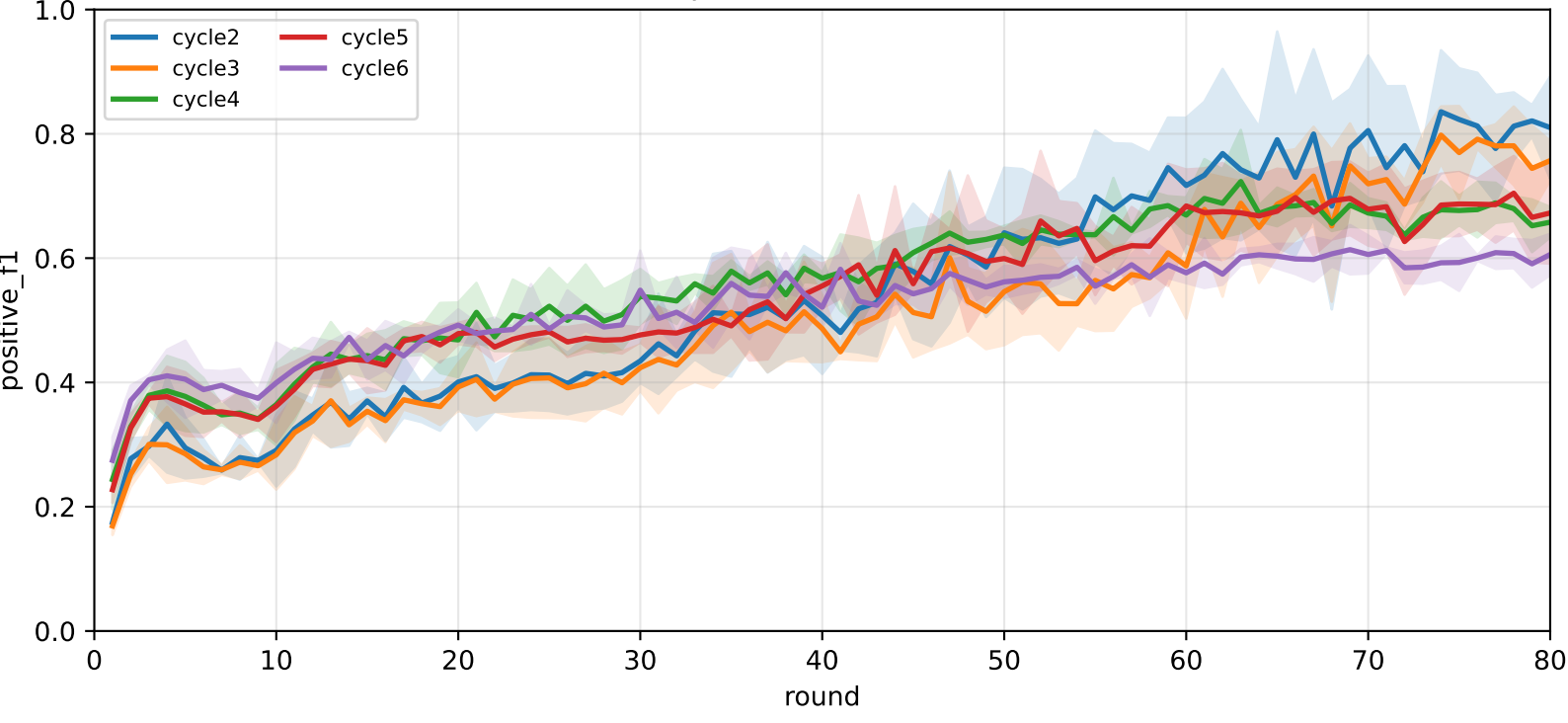
Pooled GCFL+ positive F1



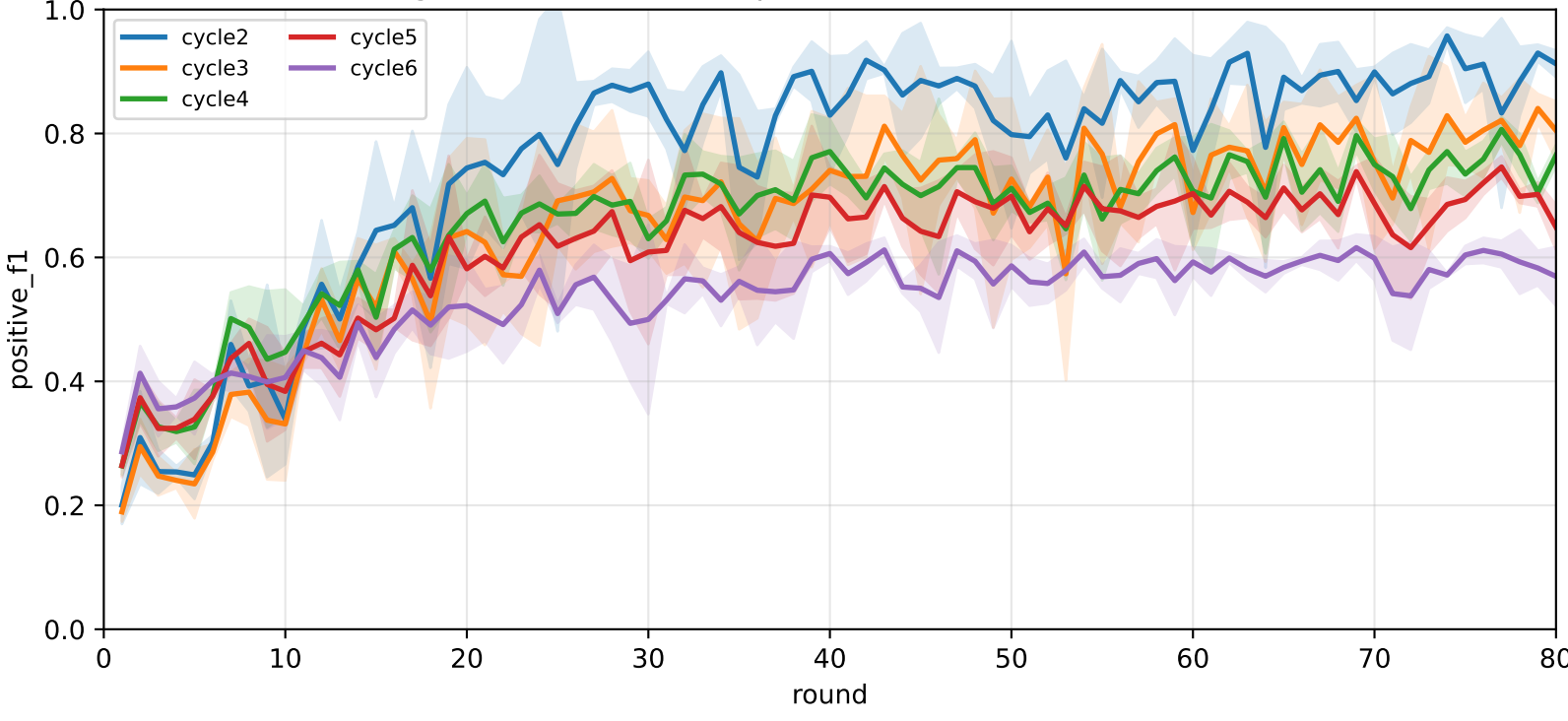
Pooled APPLE-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-Backbone-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-FedAvgBackbone-TaskHead positive F1 (after-local selected model, rounds=80)



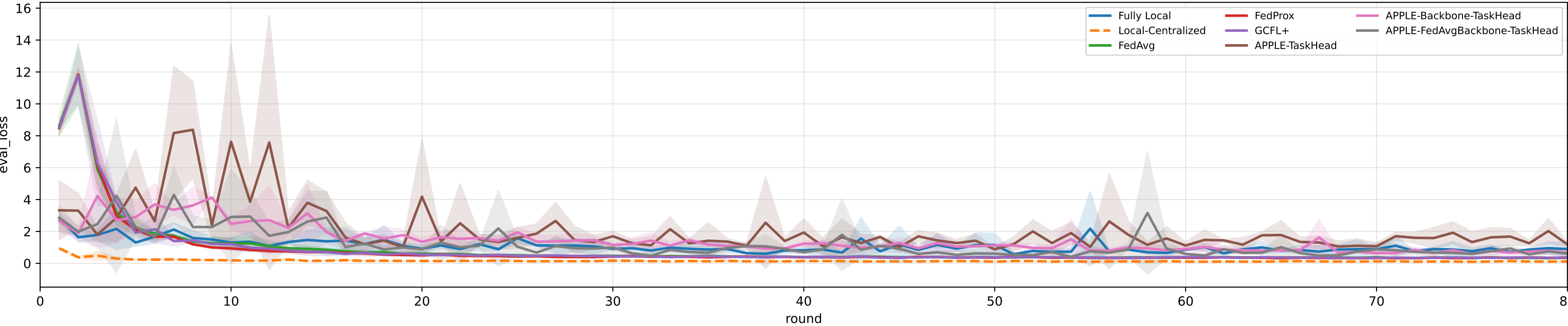
Controlled q-label heterogeneity: Mid | family aggregated / pooled performance |
family=q task_cycle6 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% |
cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

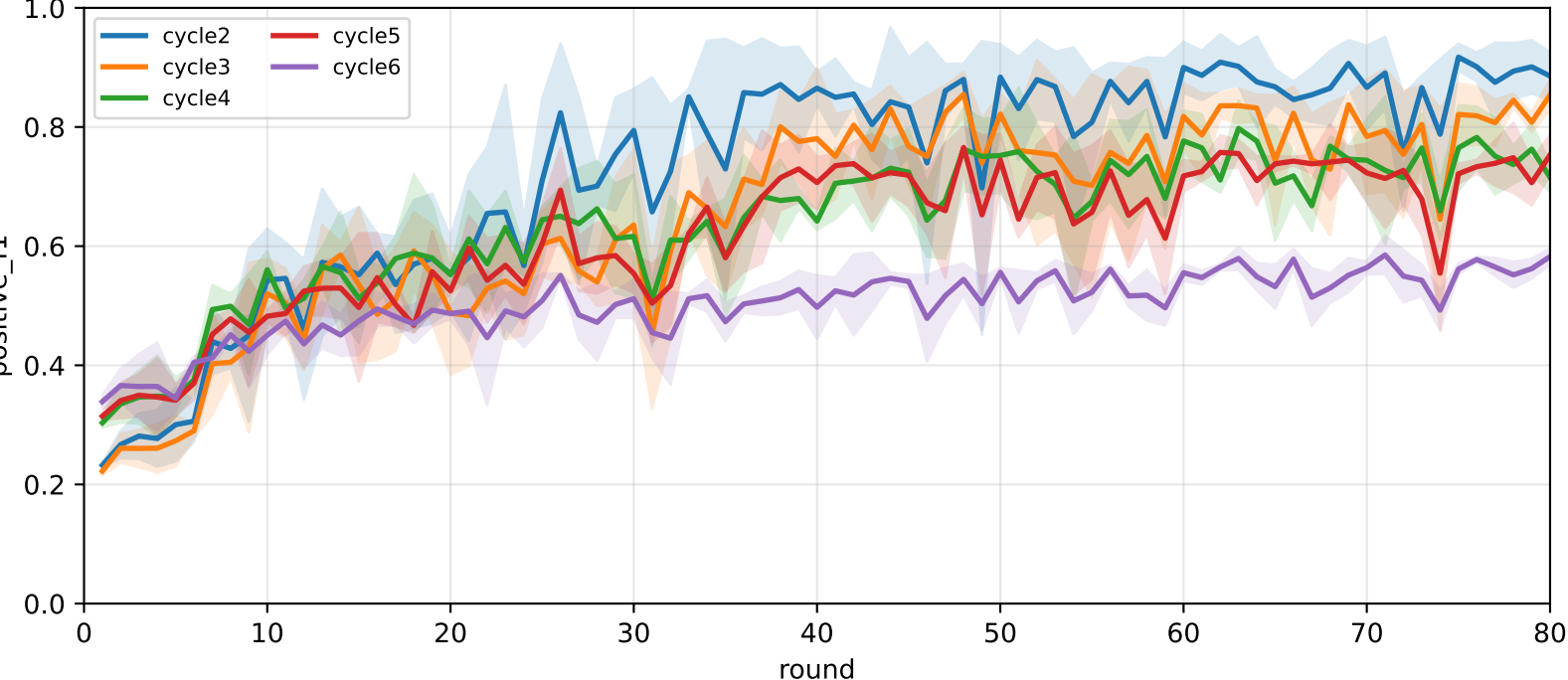
Best logged test performance (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	64.1% ± 1.8%	67.8% ± 2.4%	81.8% ± 8.2%	78.3% ± 6.8%	64.9% ± 1.4%	64.9% ± 4.4%	49.0% ± 3.2%
Local-Centralized	84.3% ± 1.6%	86.3% ± 1.4%	96.7% ± 1.1%	94.7% ± 0.6%	87.4% ± 4.2%	80.9% ± 2.2%	71.8% ± 3.9%
FedAvg	<u>70.2% ± 2.0%</u>	<u>72.8% ± 1.6%</u>	86.5% ± 4.6%	81.6% ± 4.4%	67.6% ± 2.3%	<u>70.2% ± 4.4%</u>	<u>58.0% ± 2.4%</u>
FedProx	69.0% ± 0.3%	72.0% ± 0.6%	<u>89.0% ± 2.9%</u>	79.3% ± 2.0%	<u>70.7% ± 2.4%</u>	64.4% ± 2.1%	56.4% ± 1.9%
GCFL+	69.9% ± 2.5%	72.3% ± 2.4%	84.3% ± 3.3%	<u>82.6% ± 3.5%</u>	68.4% ± 2.2%	68.3% ± 4.0%	57.8% ± 2.3%
APPLE-TaskHead	56.3% ± 1.0%	58.1% ± 2.0%	67.6% ± 10.3%	59.2% ± 1.6%	60.4% ± 0.8%	56.5% ± 3.8%	47.0% ± 0.5%
APPLE-Backbone-TaskHead	64.7% ± 2.0%	66.2% ± 2.8%	71.9% ± 8.9%	71.9% ± 6.5%	64.5% ± 1.8%	66.6% ± 5.1%	55.9% ± 1.3%
APPLE-FedAvgBackbone-TaskHead	66.5% ± 2.4%	69.6% ± 2.0%	87.8% ± 4.0%	69.9% ± 5.1%	67.4% ± 3.0%	<u>70.2% ± 2.0%</u>	52.8% ± 2.6%

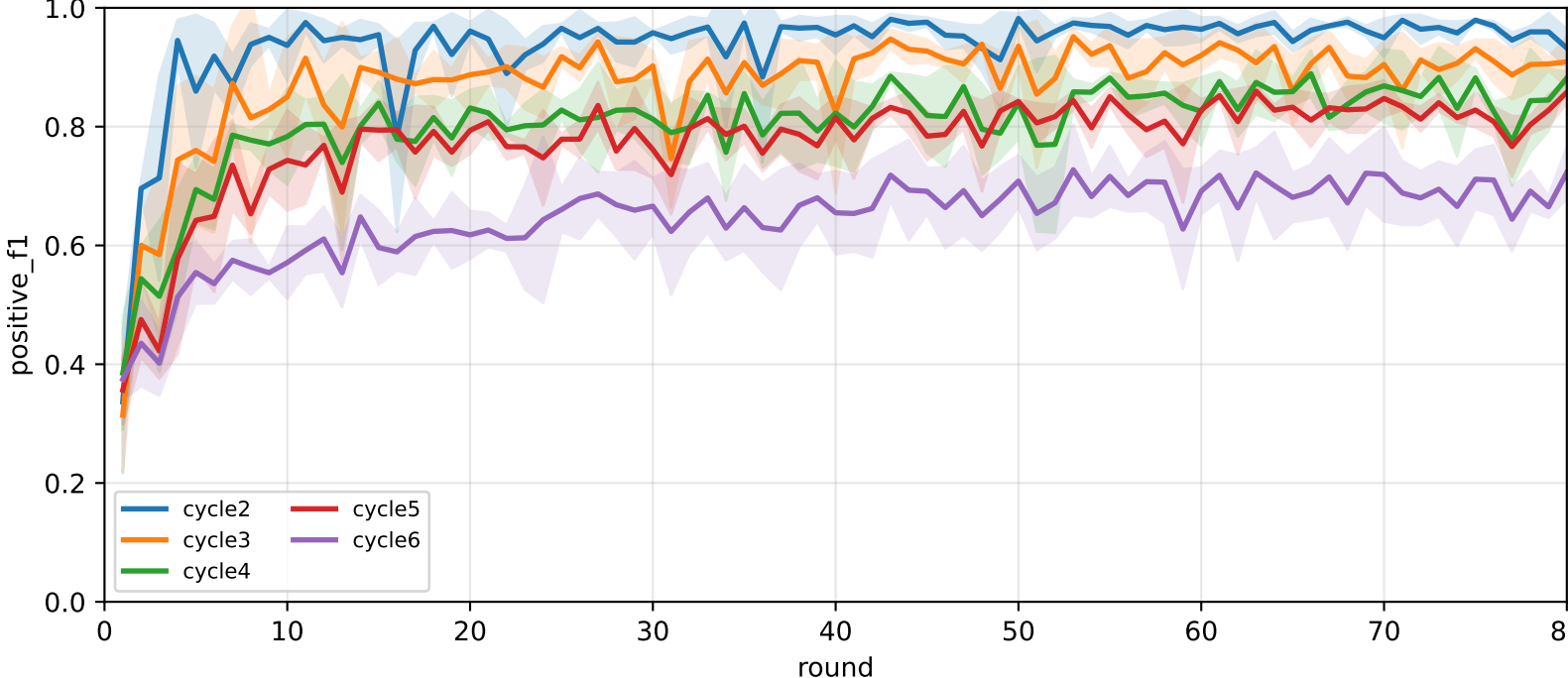
Validation loss comparison (APPLE family: rounds=80, dr_lr=0.001, mu=0.001, Lfrac=0.2)



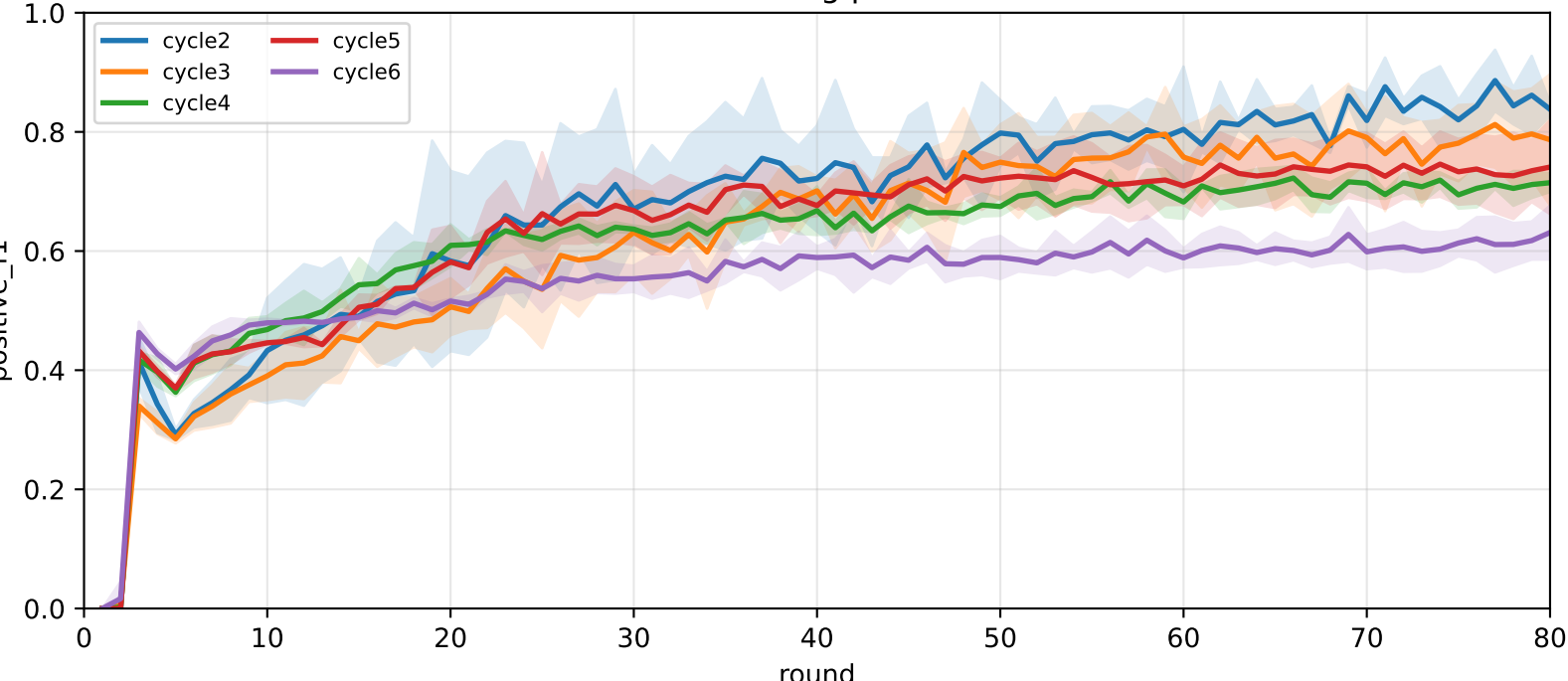
Pooled Fully Local positive F1



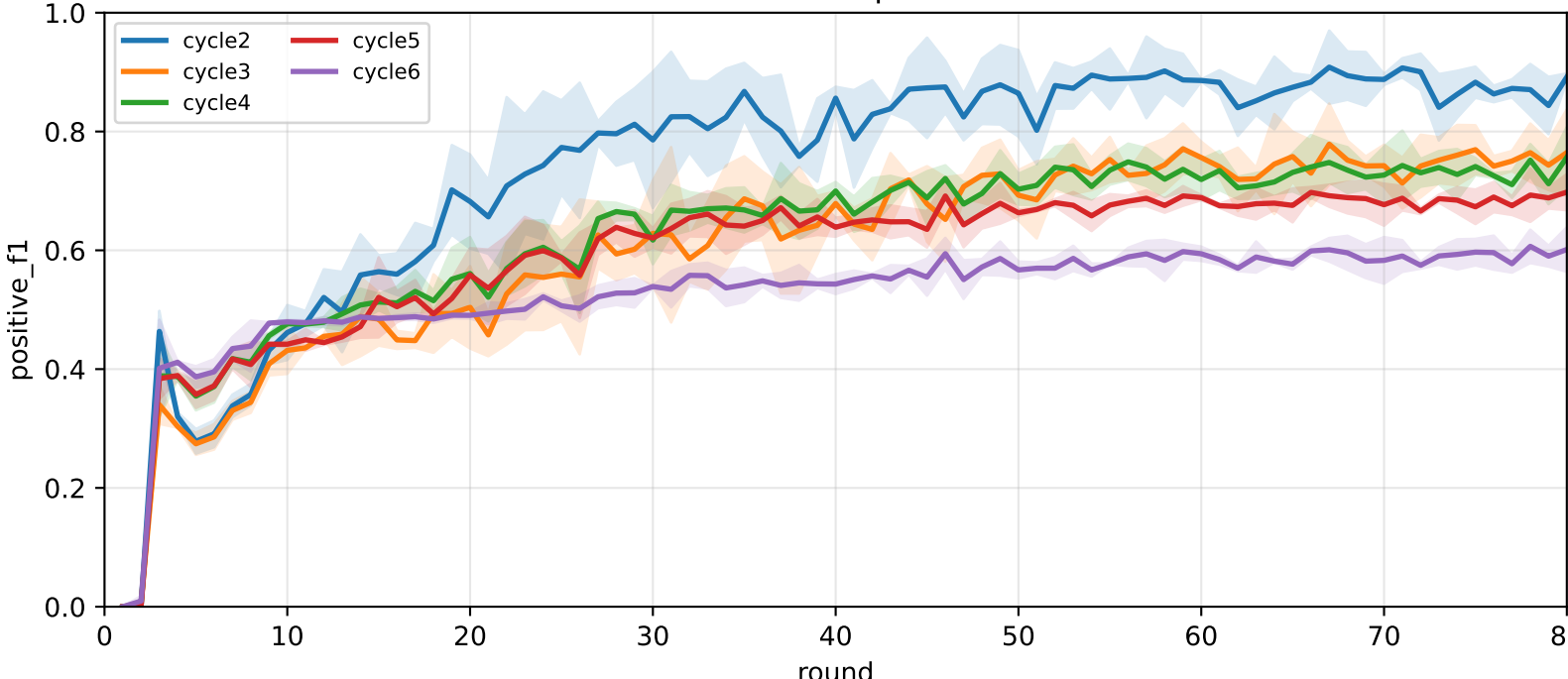
Pooled Local-Centralized positive F1



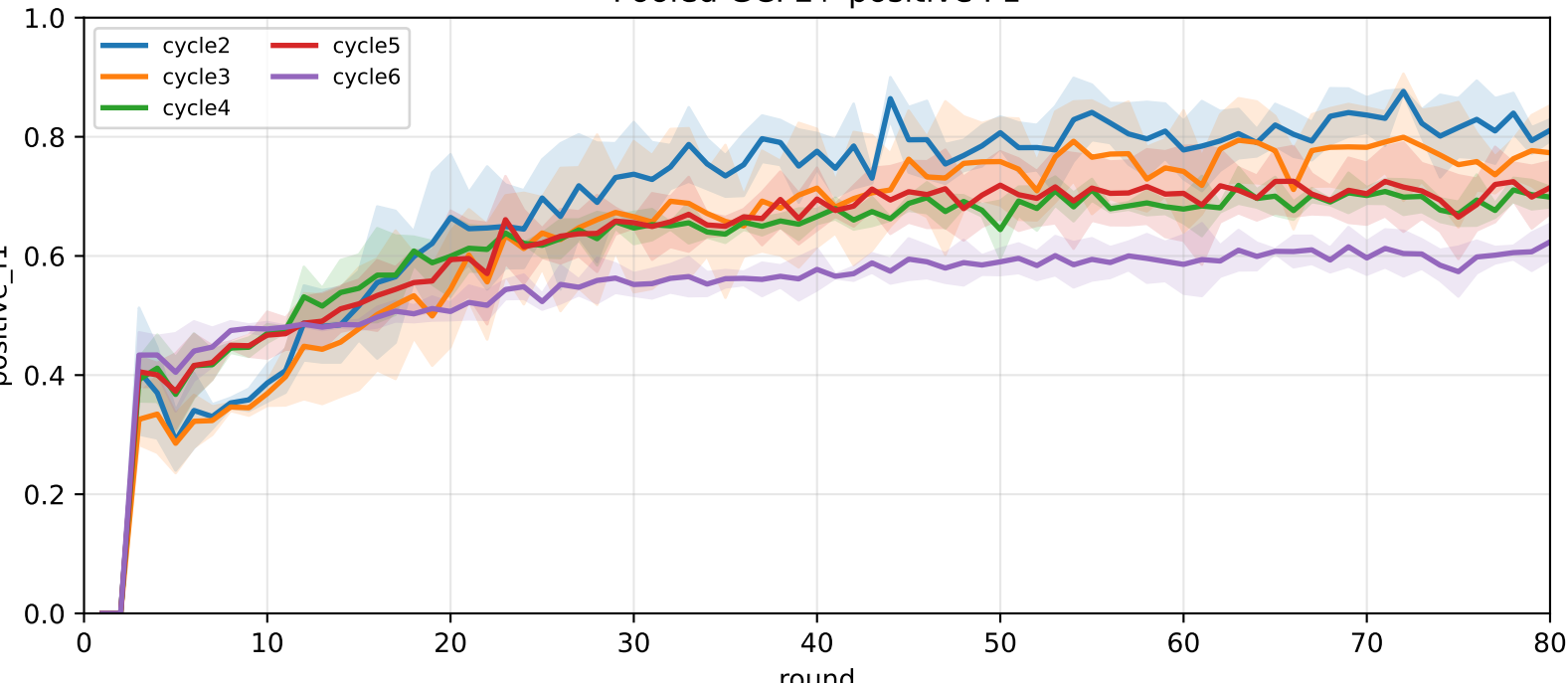
Pooled FedAvg positive F1



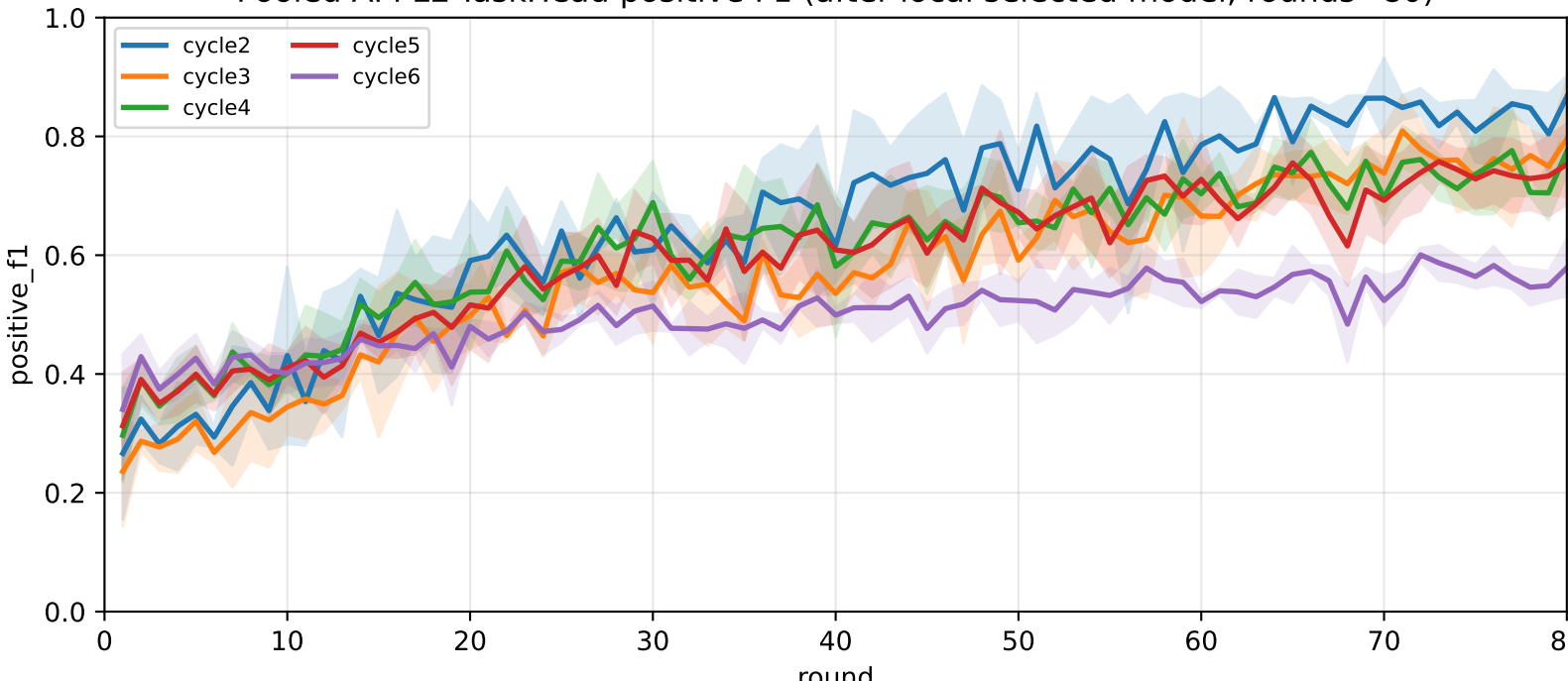
Pooled FedProx positive F1



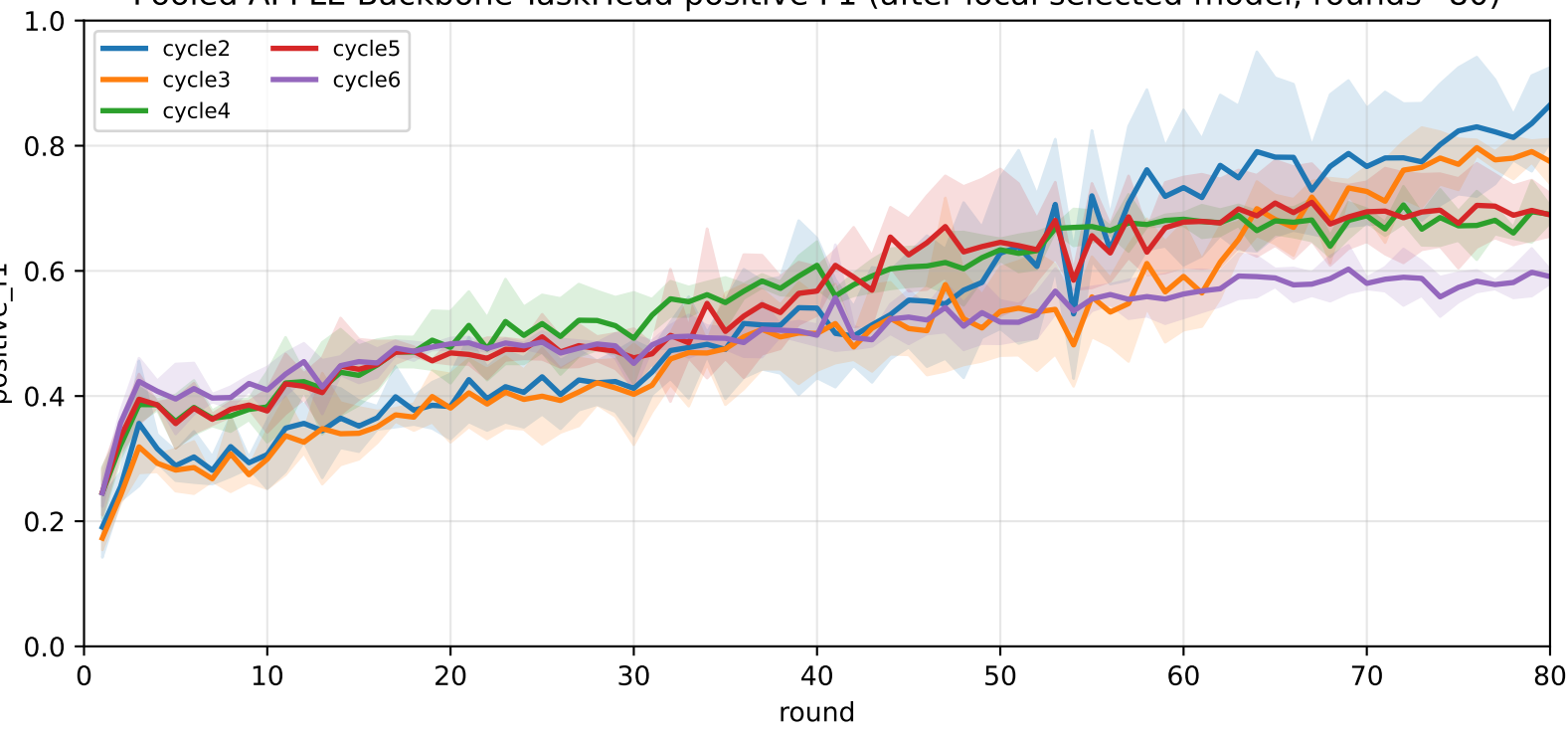
Pooled GCFL+ positive F1



Pooled APPLE-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-Backbone-TaskHead positive F1 (after-local selected model, rounds=80)



Pooled APPLE-FedAvgBackbone-TaskHead positive F1 (after-local selected model, rounds=80)

